

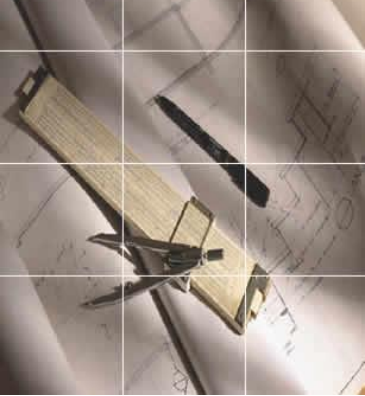
Using Drawing Tools & Applied Geometry



TOPICS

- Preparation of Tools.
- Using of Tools
- Applied Geometry

Preparation of Tools



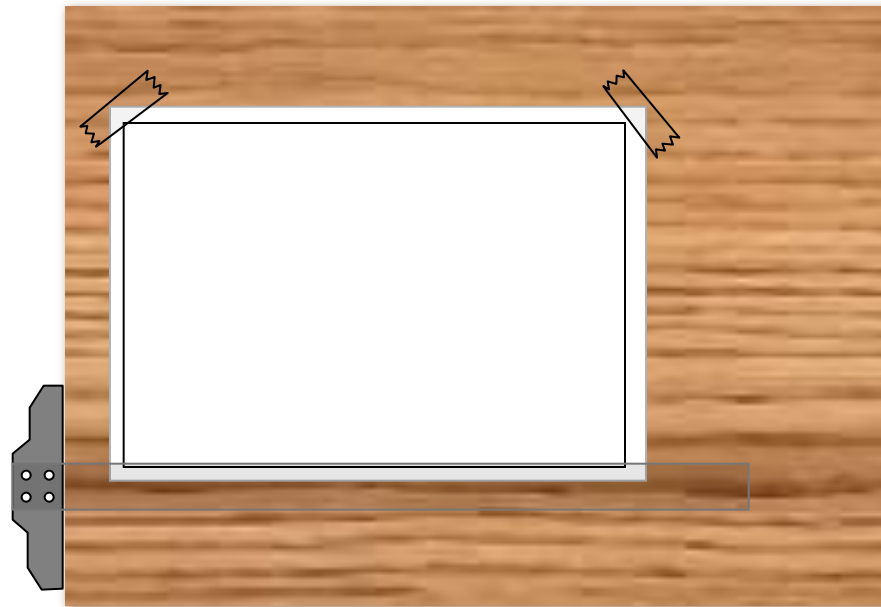
Fastening Paper to Drafting Board

1. Place the paper close to the table's left edge.
2. Move the paper until its lower edge place about the top edge of T-square.



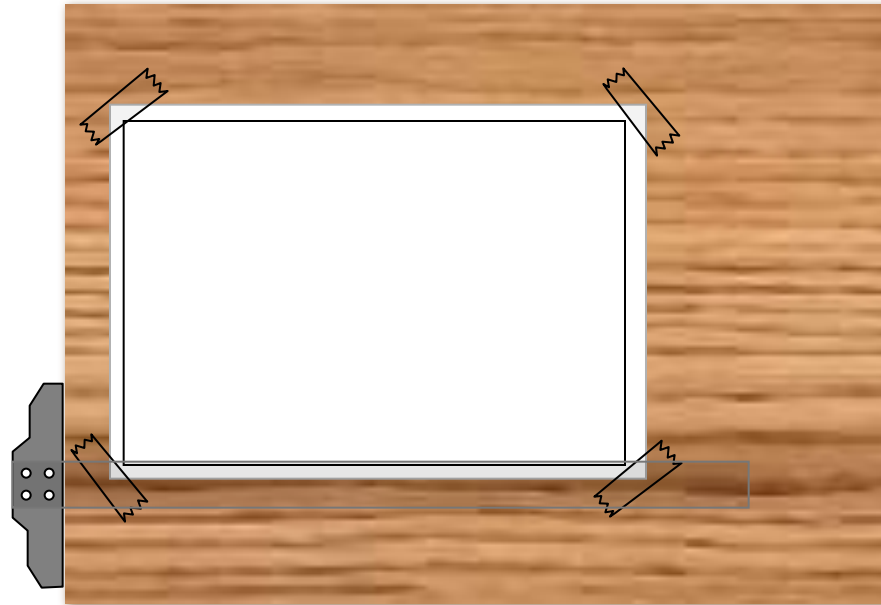
Fastening Paper to Drafting Board

3. Align the top edge of the paper with T-square blade.
4. Attach the paper's corners with tape.



Fastening Paper to Drafting Board

5. Move T-square down to smooth the paper.
6. Attach the remaining paper's corners with tape.



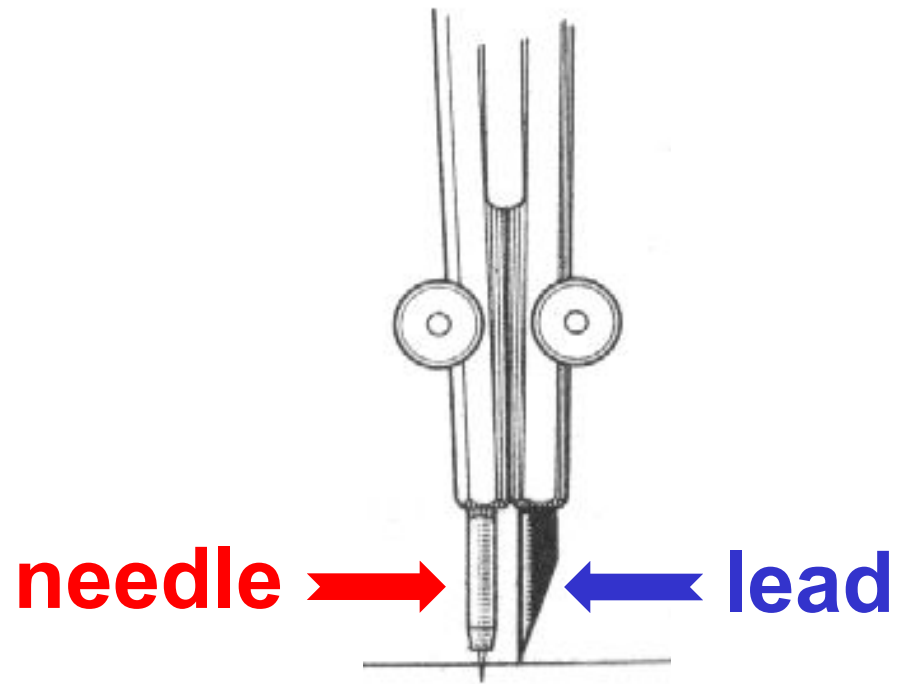
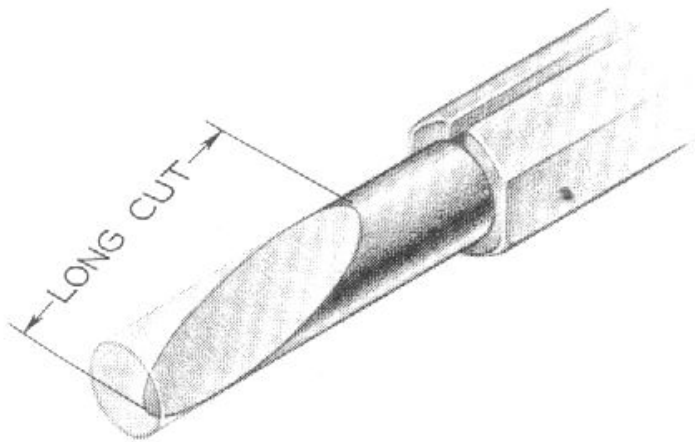
Sharpening the Pencil

1. Remove the wood with penknife while expose a lead about 8-10 mm.
2. Polish the lead into a conical shape with a sandpaper.
3. Clean the lead with tissue paper.



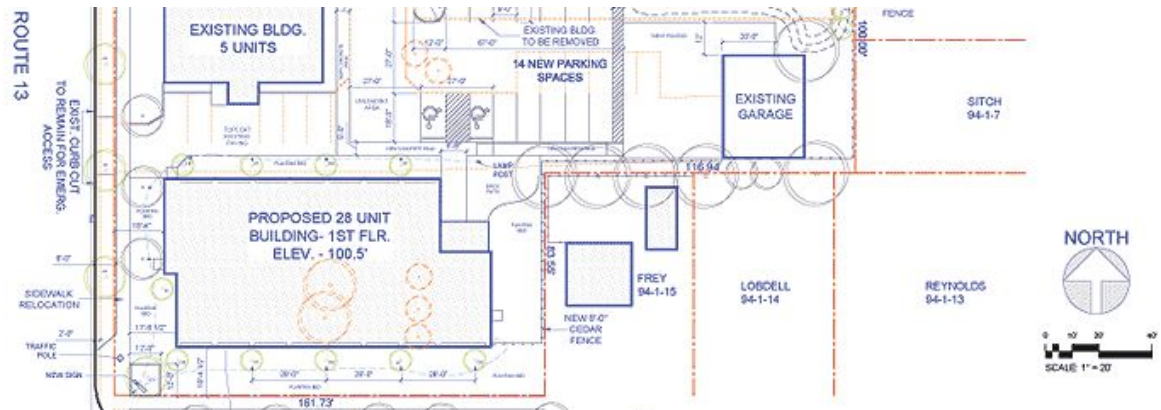
Preparing the Compass

1. Sharpen the lead with a sandpaper.
2. Adjust the **needle** and the **lead** so that the tip of the needle extends slightly more than the lead.





Using the Tools



Function of the Tools

Tools

1. T-square
2. Triangles



Straight line

3. Compass
4. Circle template

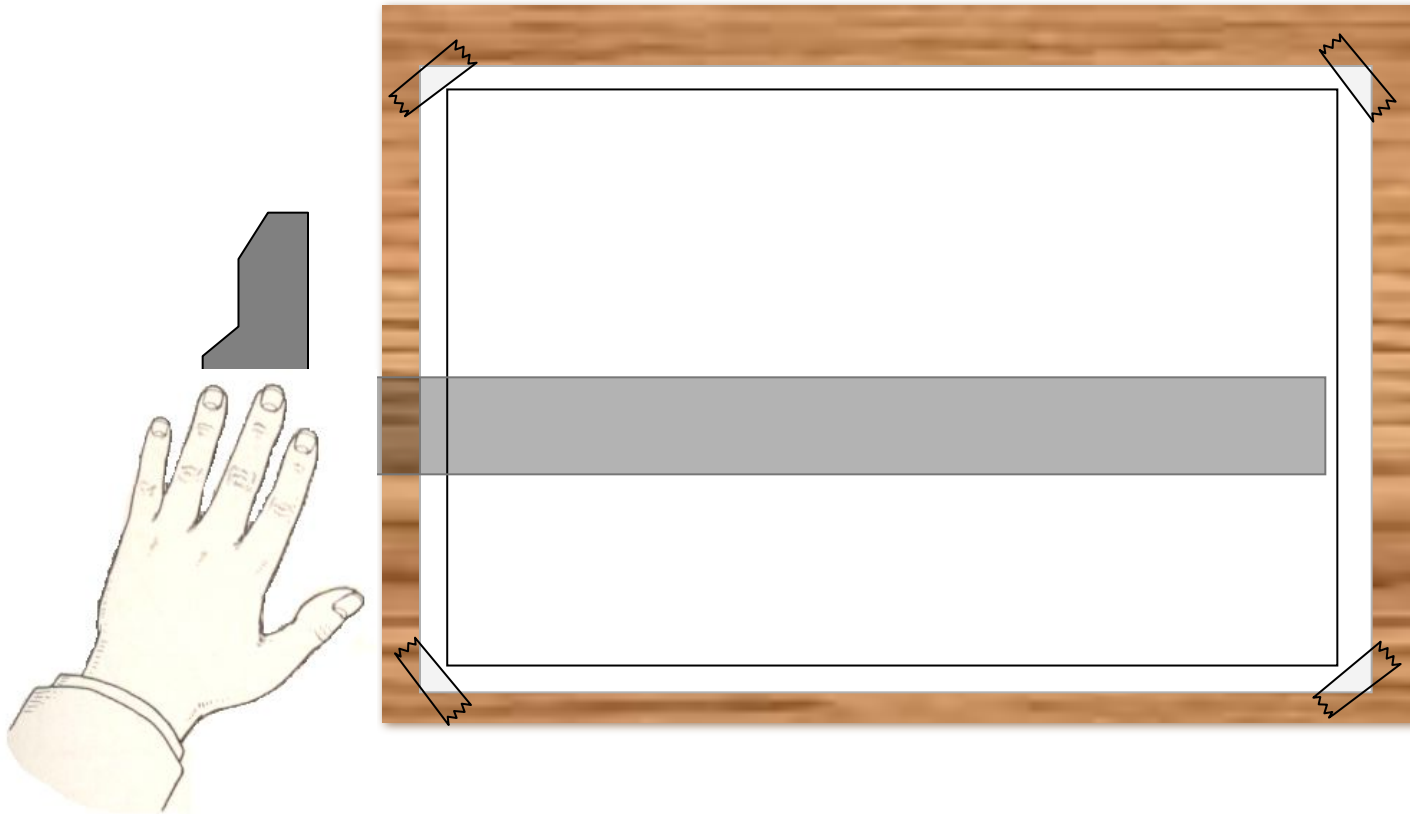


Arc, Circle

Shape to be drawn

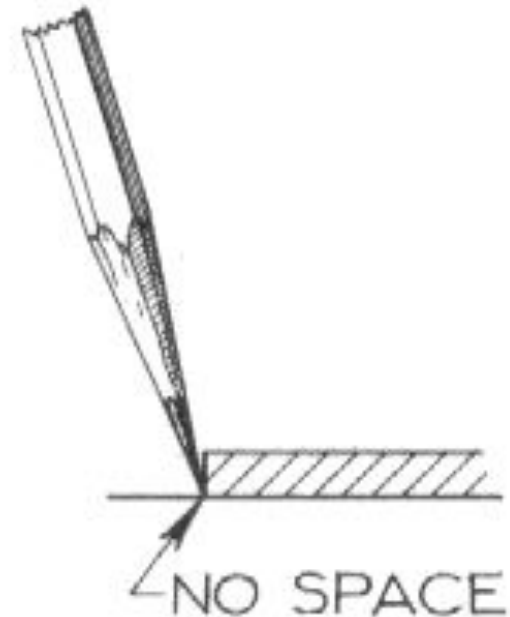
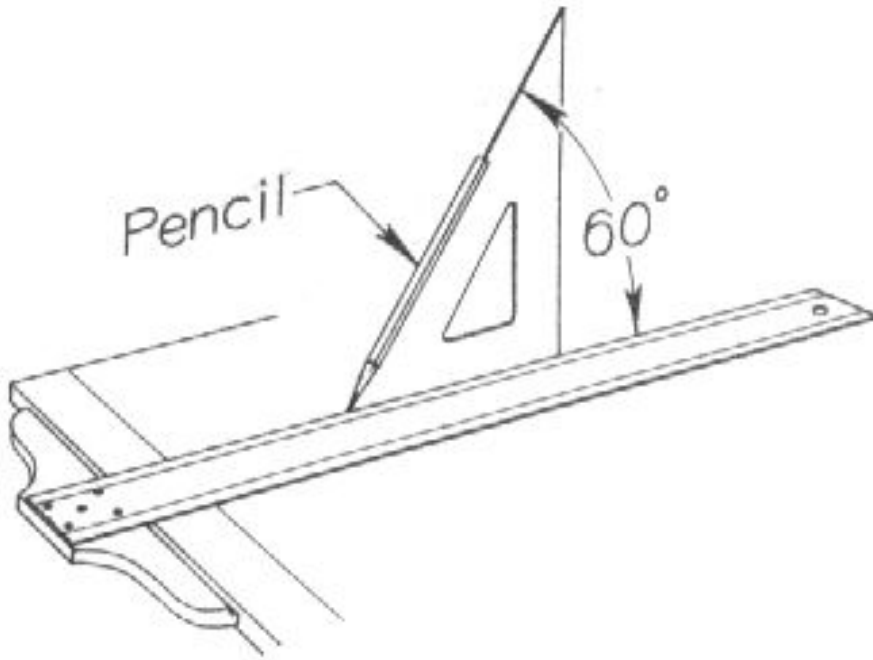
Draw a Horizontal Line

1. Press the T-square head against the left edge of the table.
2. Smooth the blade to the right.



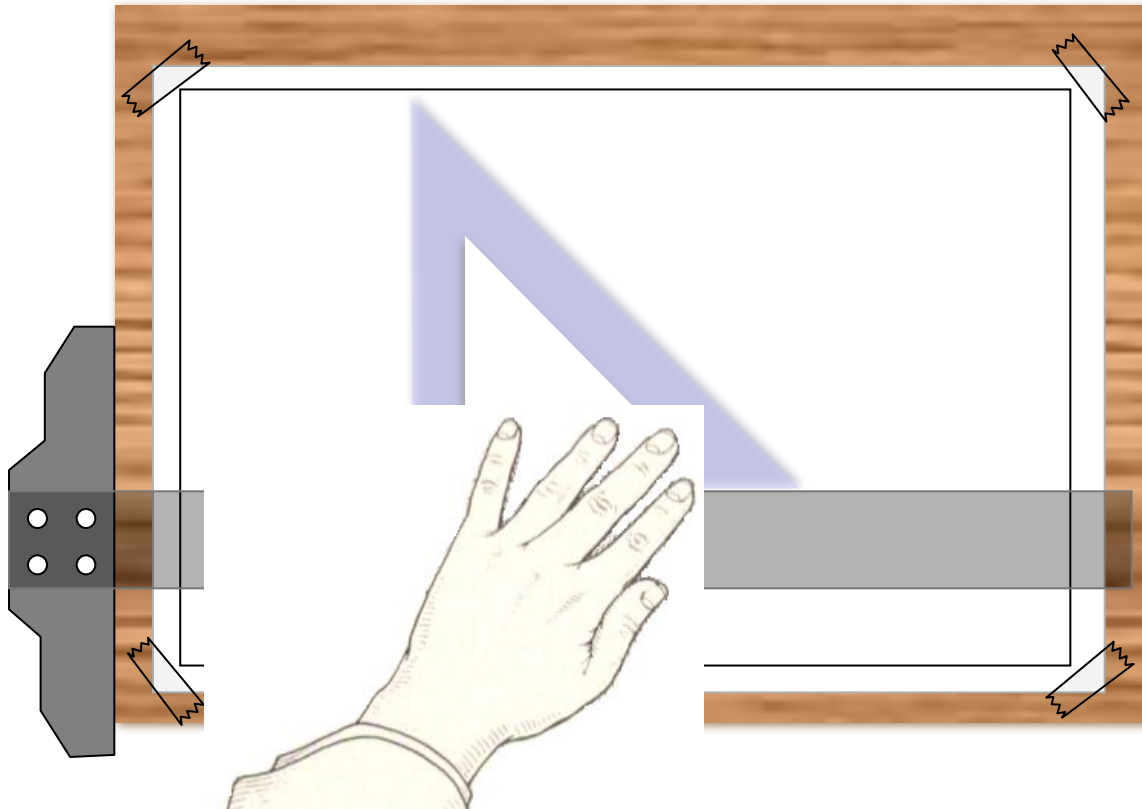
Draw a Horizontal Line

3. Lean the pencil at an angle about 60° with the paper in the direction of the line and slightly “toed in”.
4. Draw the line from left to right while **rotating the pencil slowly**.



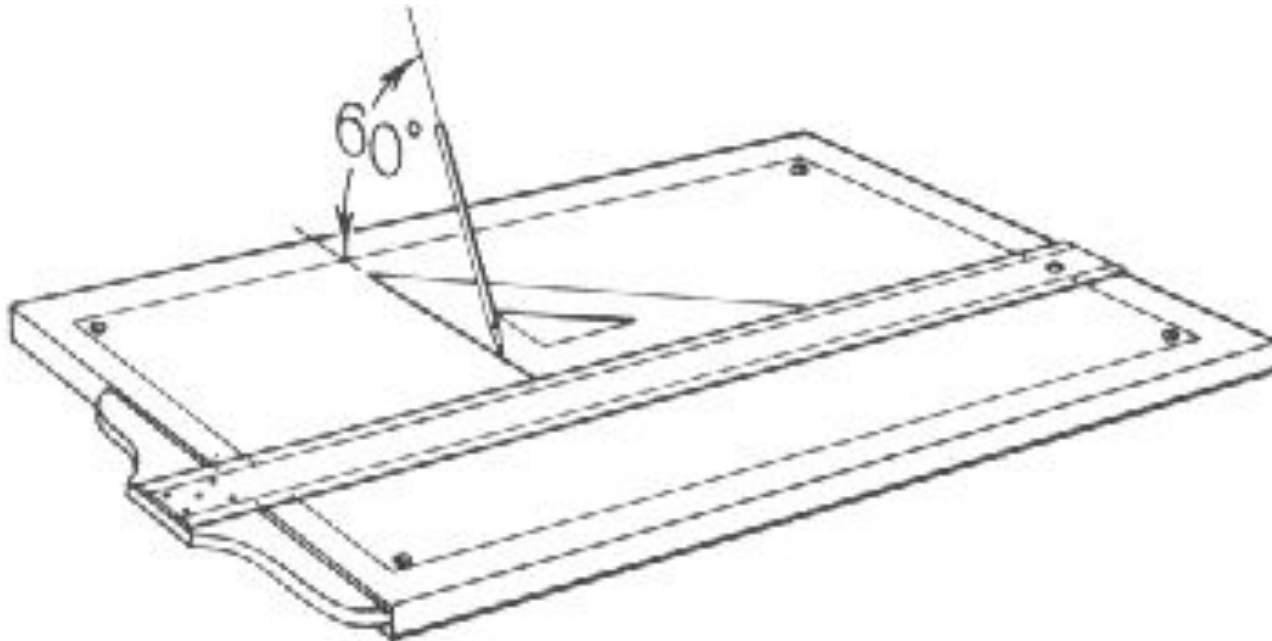
Draw a Vertical Line

1. Set T-square as before. Place any triangle on T-square edge.
2. Slide your left hand to hold both T-square and triangle in position.



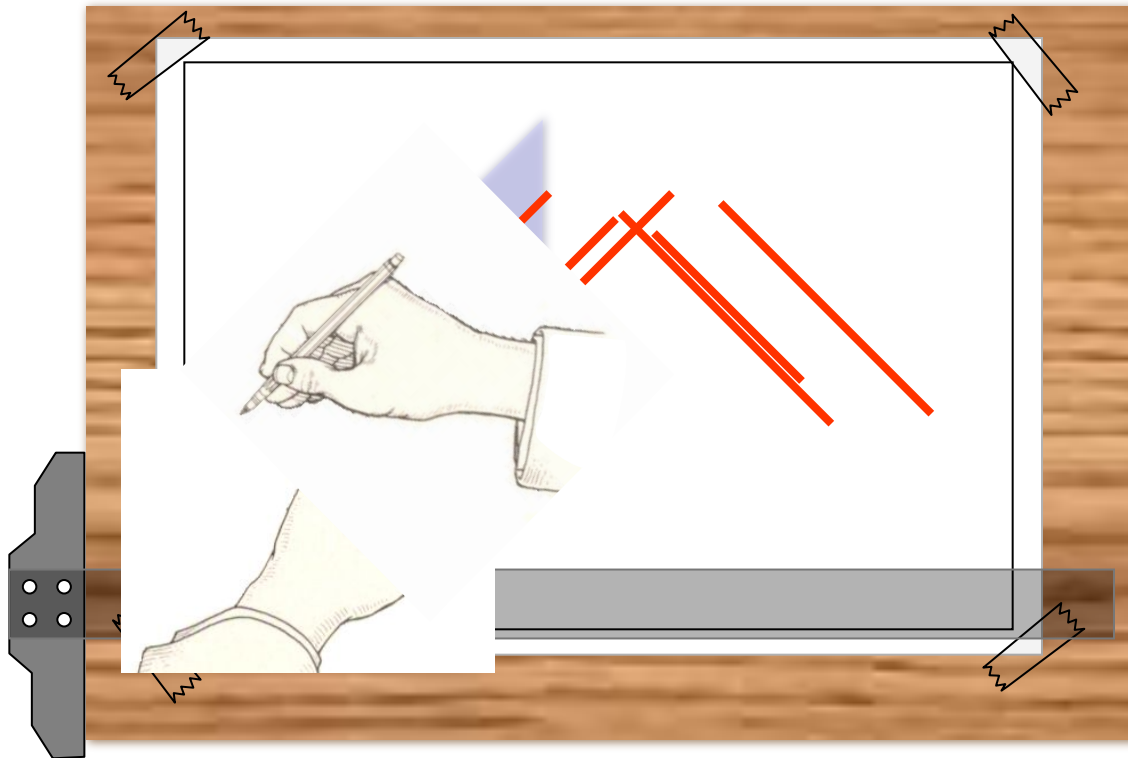
Draw a Vertical Line

3. Lean the pencil to the triangle.
4. Draw the line upward while rotating the pencil slowly.



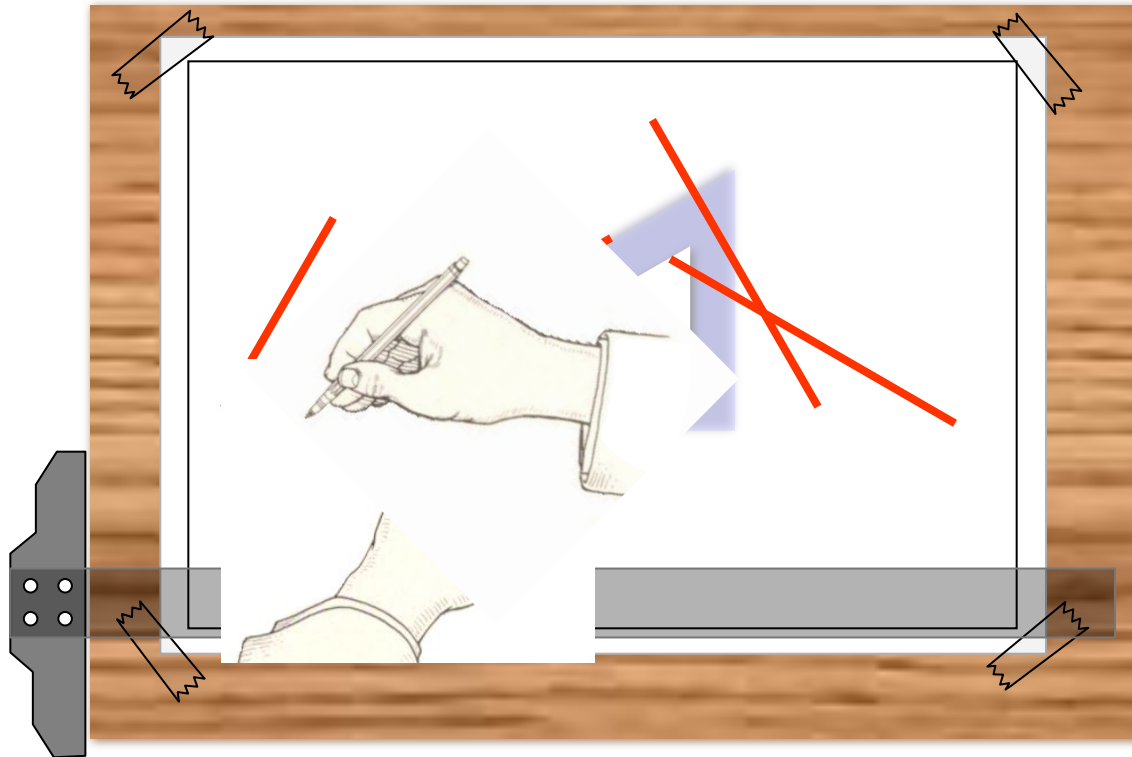
Draw a line at 45° with horizontal

1. Place 45° triangle on the T-square edge and press them firmly against the paper.
2. Draw the line in the direction as shown below.



Draw a line at angle 30° and 60°

1. Place 30°-60° triangle on the T-square edge and press them firmly against the paper.
2. Draw the line in the direction as shown below.



Draw the lines at 15° increment

0 deg.

15 deg. = $-30 + 45$

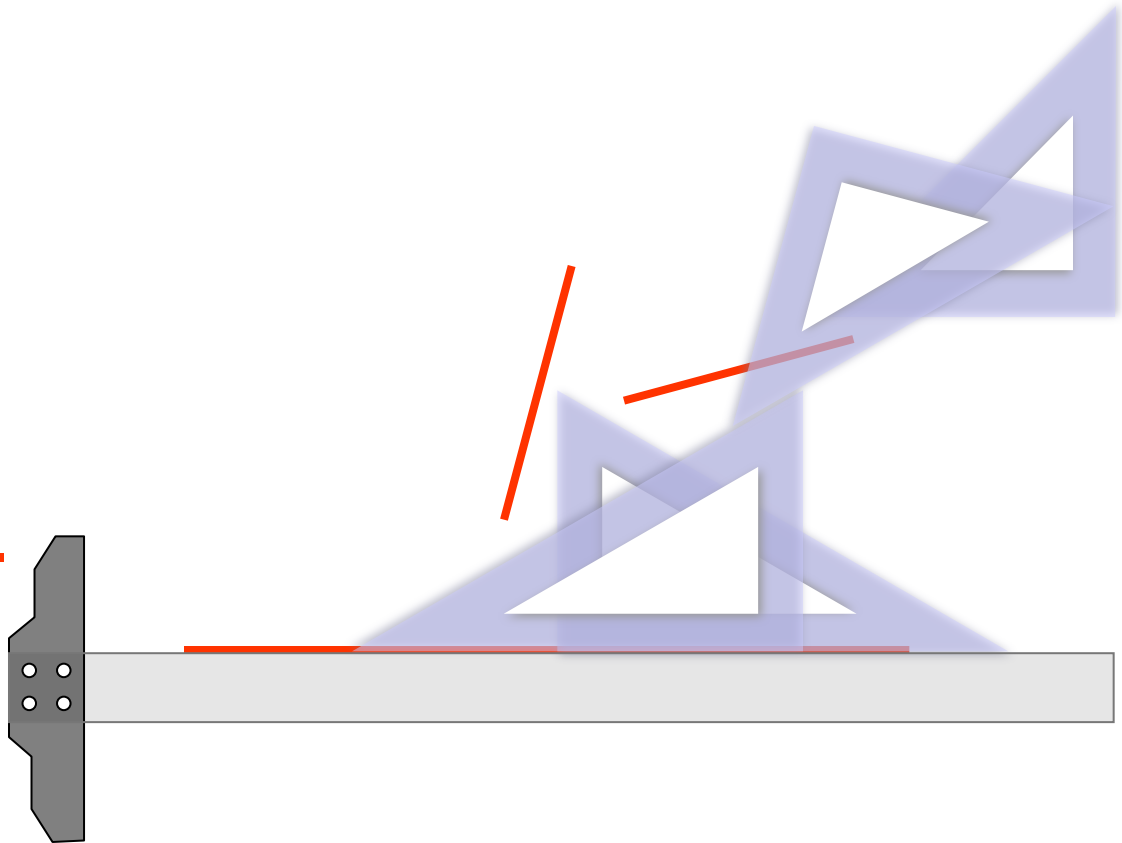
30 deg. deg

45 deg. Already demonstrated.

60 deg.

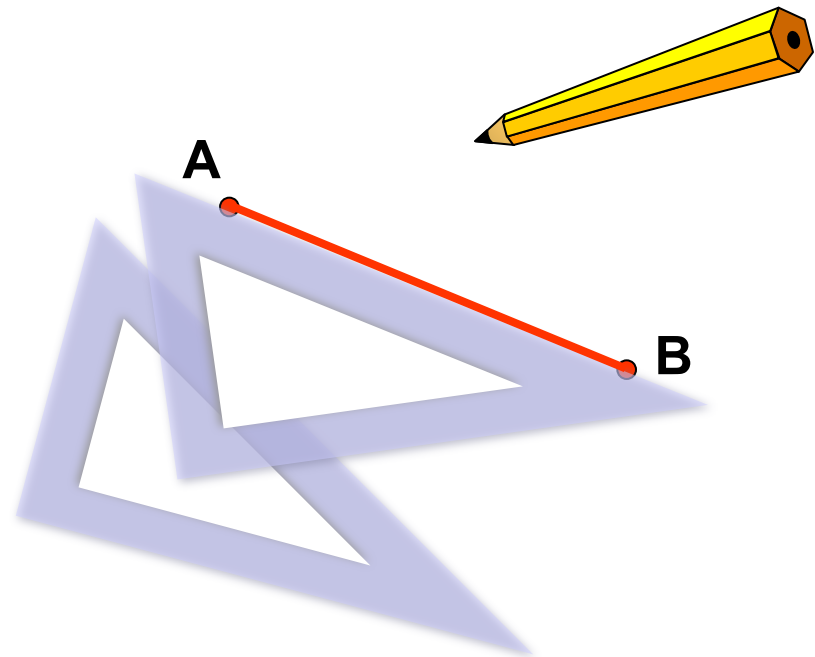
75 deg. = $30 + 45$ deg

90 deg. Already demonstrated.



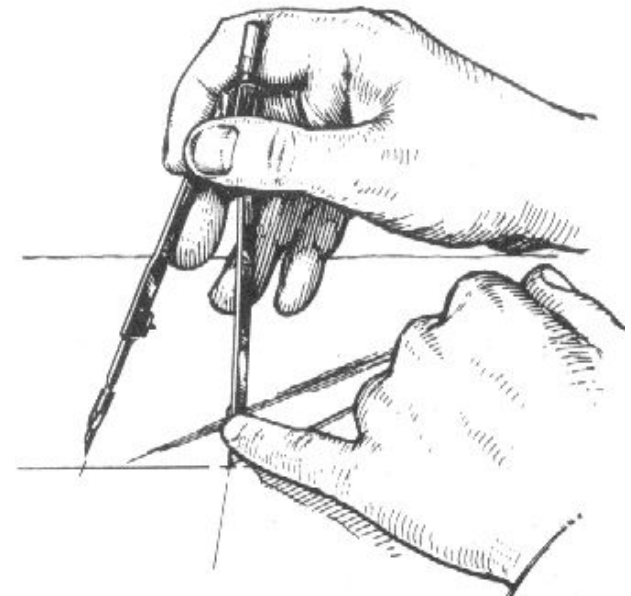
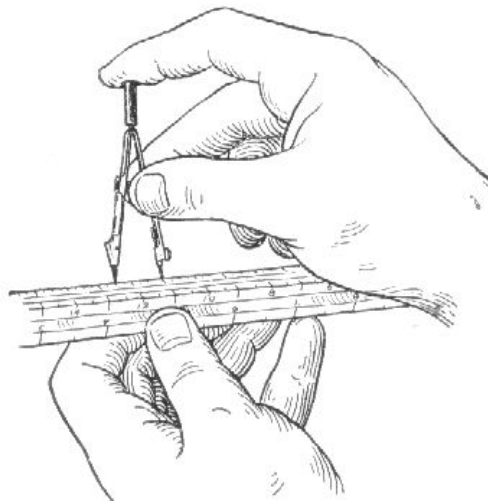
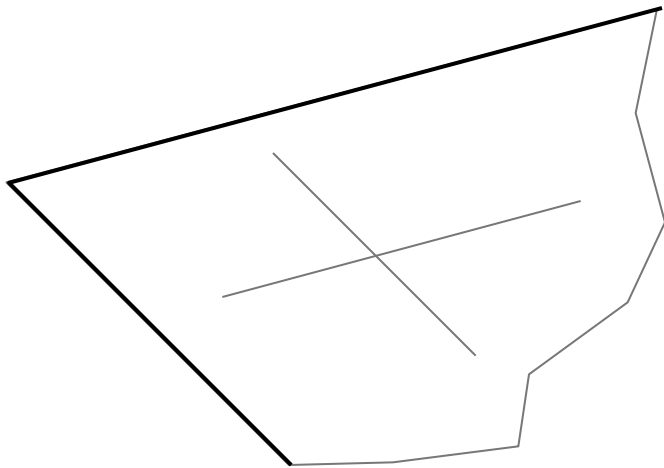
Draw the line passing through two given points

1. Place the pencil tip at one of the points.
2. Place the triangle against the pencil tip.
3. Swing the triangle around the pencil tip until its edge align with the second point.
4. Draw a line.



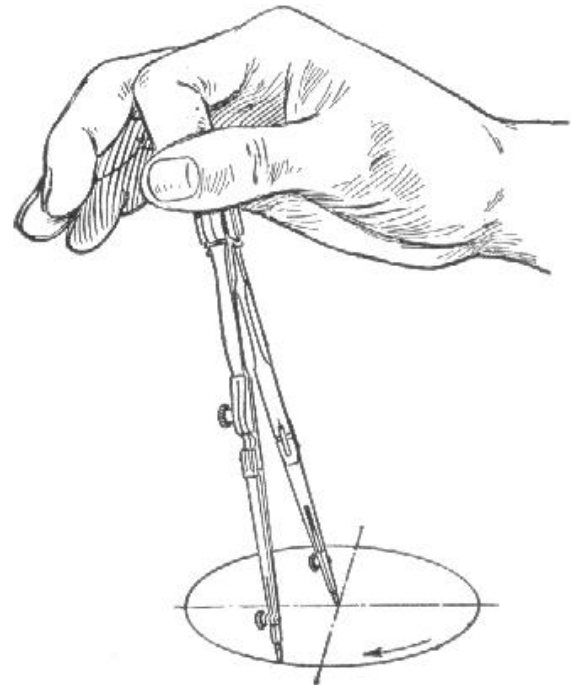
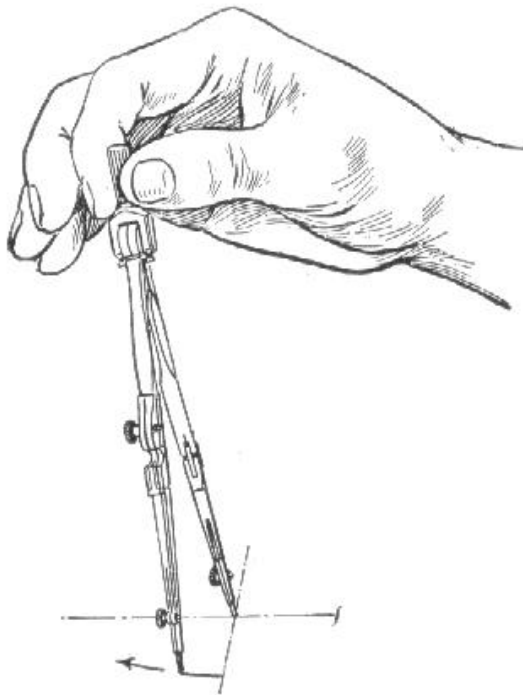
Using the Compass

1. **Locate the center** of the circle by two intersecting lines.
2. Adjust the distance between needle and lead to a distance equal to radius of the circle.
3. Set the needle point at center.



Using the Compass

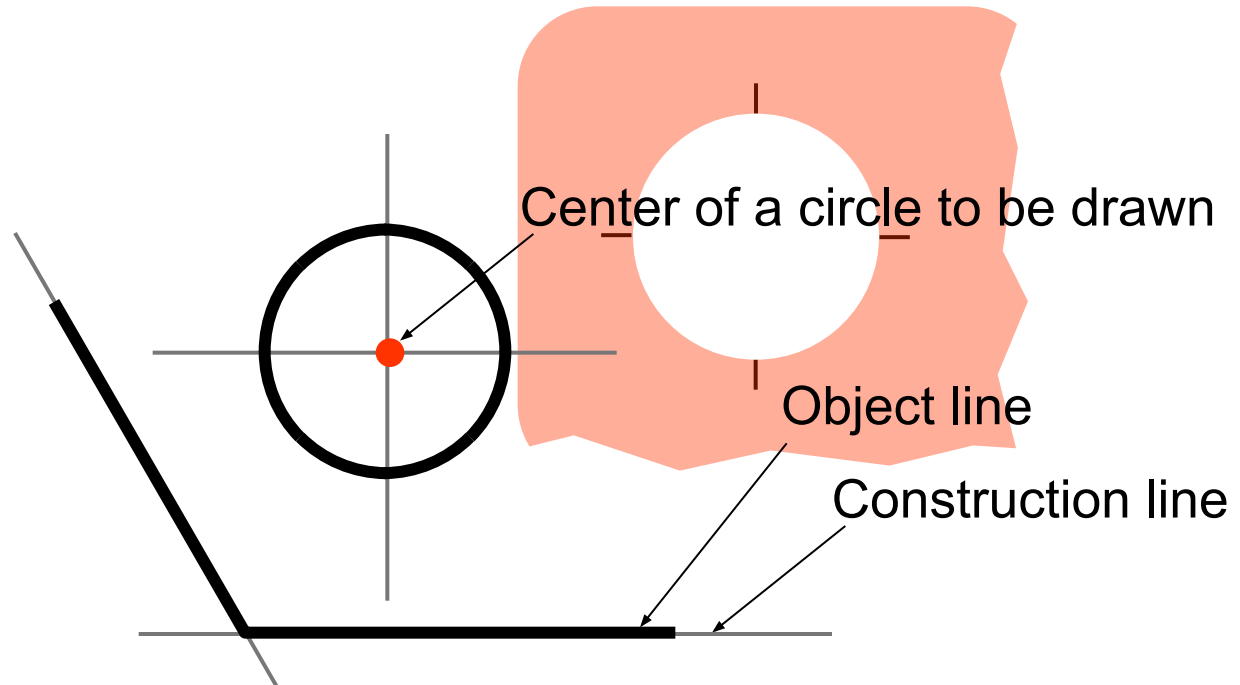
4. **Start circle.** Apply enough pressure to the needle, holding compass handle between thumb and index fingers.
5. **Complete circle.** Revolve handle clockwise.



Using a Circle Template

1. Draw two perpendicular lines that pass through center of a circle to be drawn.
2. Place the template till all marking coincide with center lines.
3. Tracing the circle. (Hold the pencil normal to the paper.)

Given



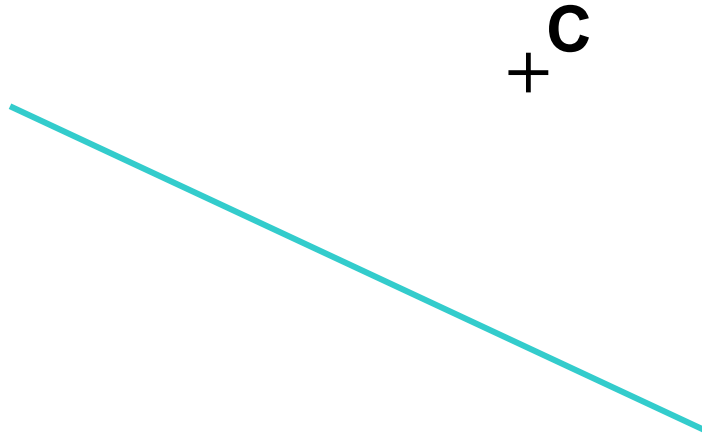


Applied Geometry



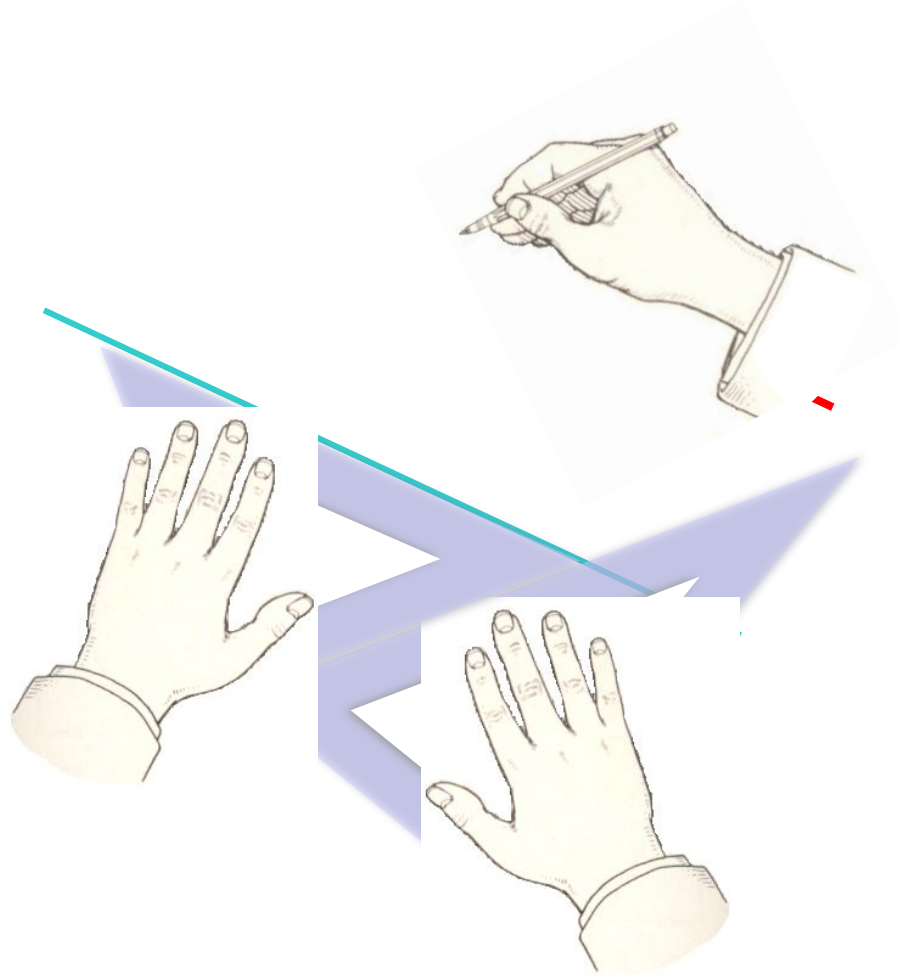
**To draw the line parallel to a given line
and passes through a given point**

Given



To draw the line parallel to a given line and passes through a given point

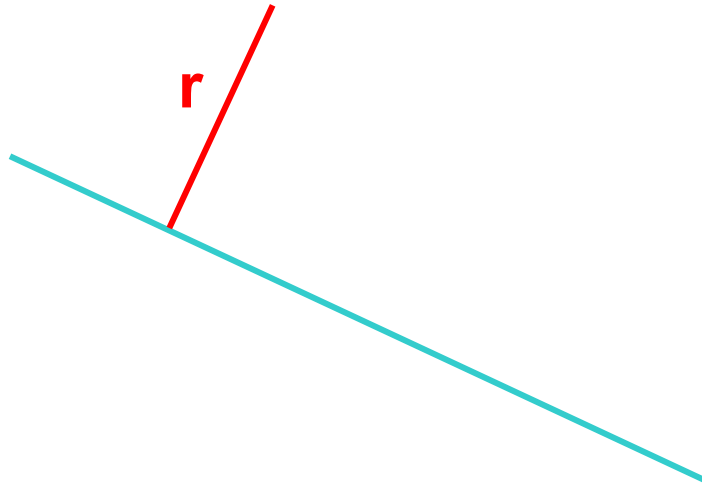
Given



Repeat

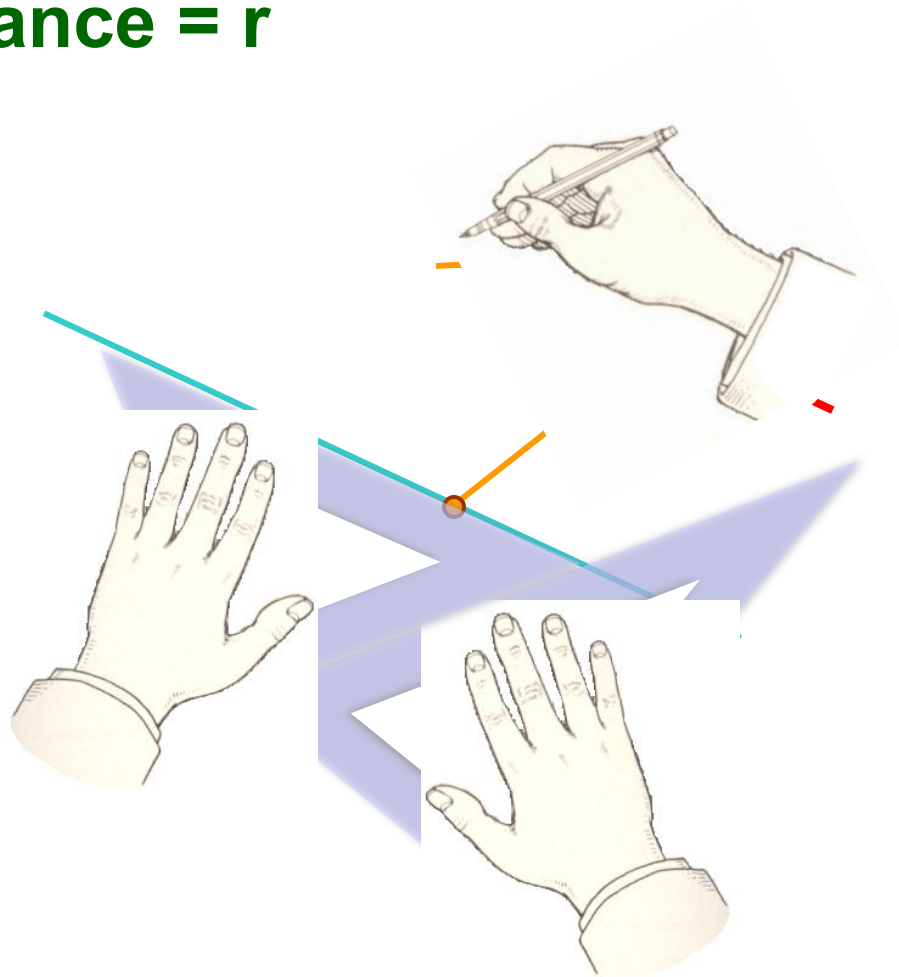
To draw the line parallel to a given line with a specified distance

Given distance = r



To draw the line parallel to a given line with a specified distance

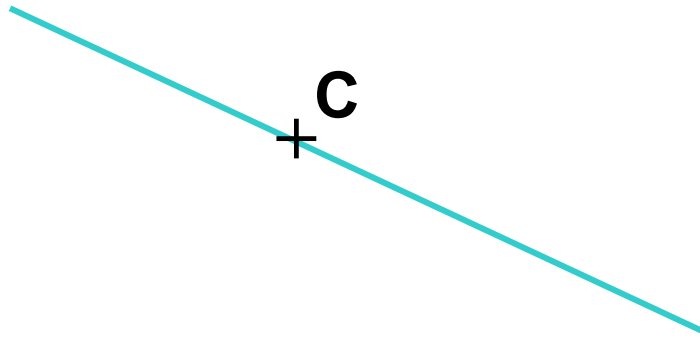
Given distance = r



Repeat

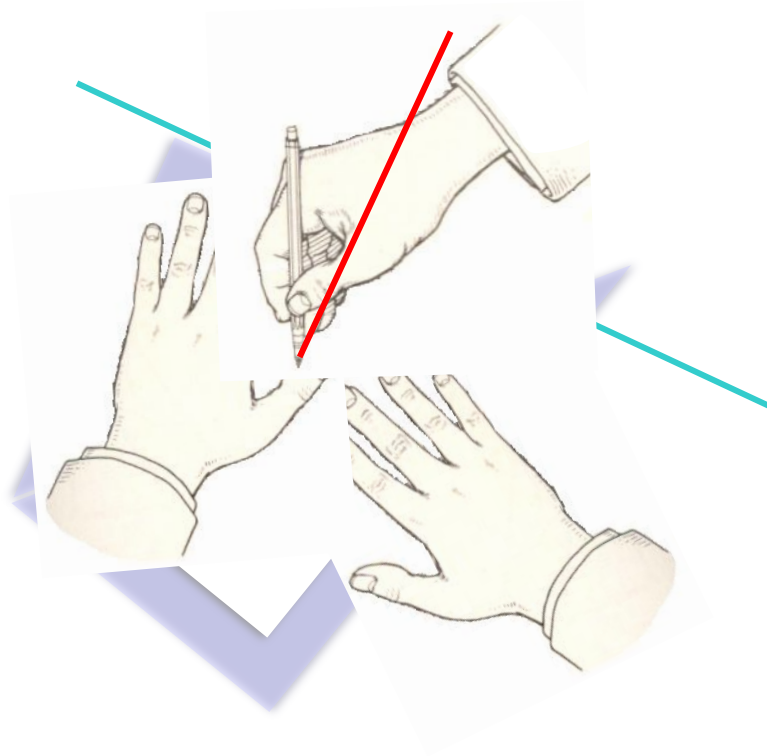
To draw the line perpendicular to a given line at a given point

Adjacent-sides method



To draw the line perpendicular to a given line at a given point

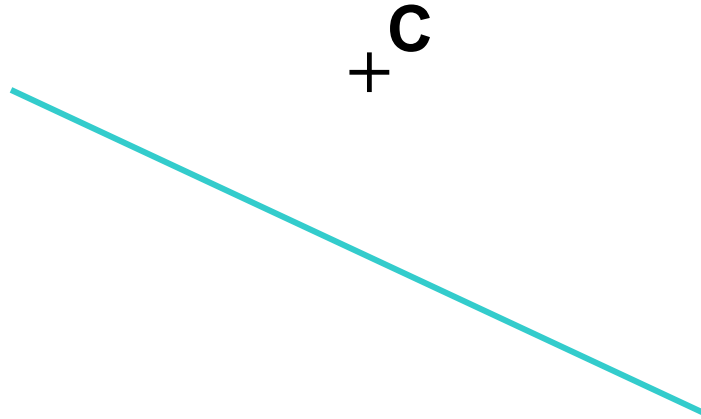
Adjacent-sides method



Repeat

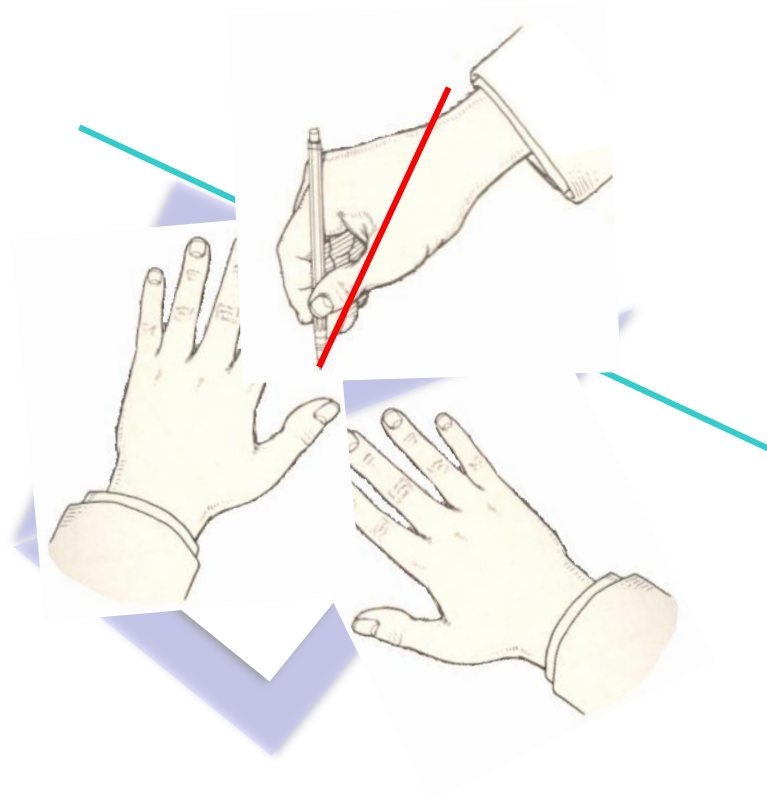
To draw the line perpendicular to a given line from a point not on the line

Adjacent-sides method



To draw the line perpendicular to a given line from a point not on the line

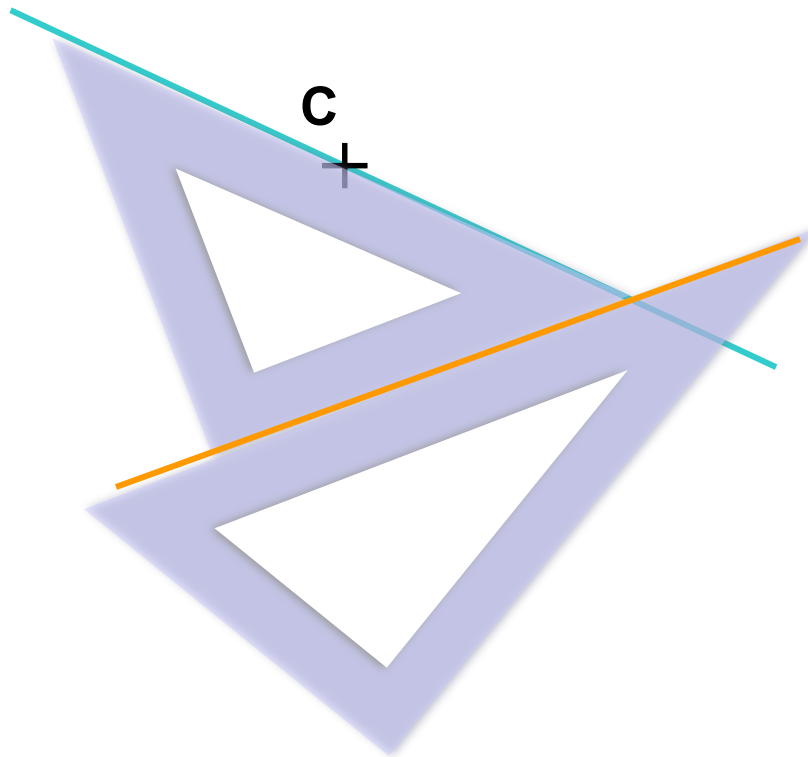
Adjacent-sides method



Repeat

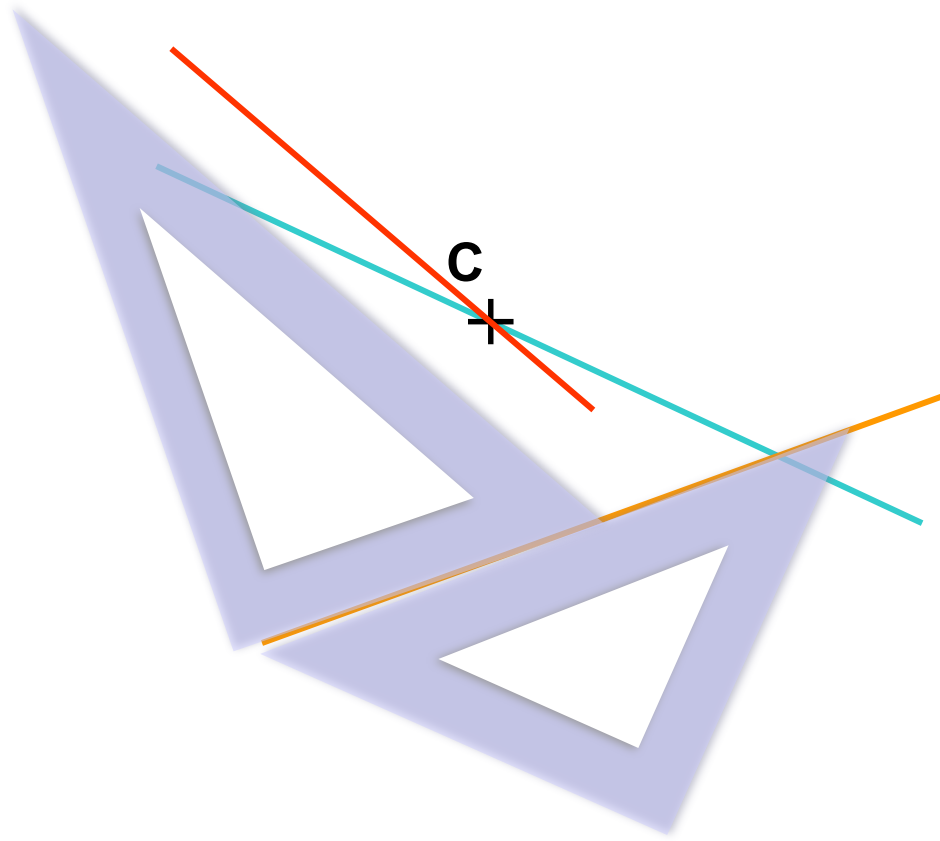
To draw a line making 15° with a given line and pass through a given point.

Given



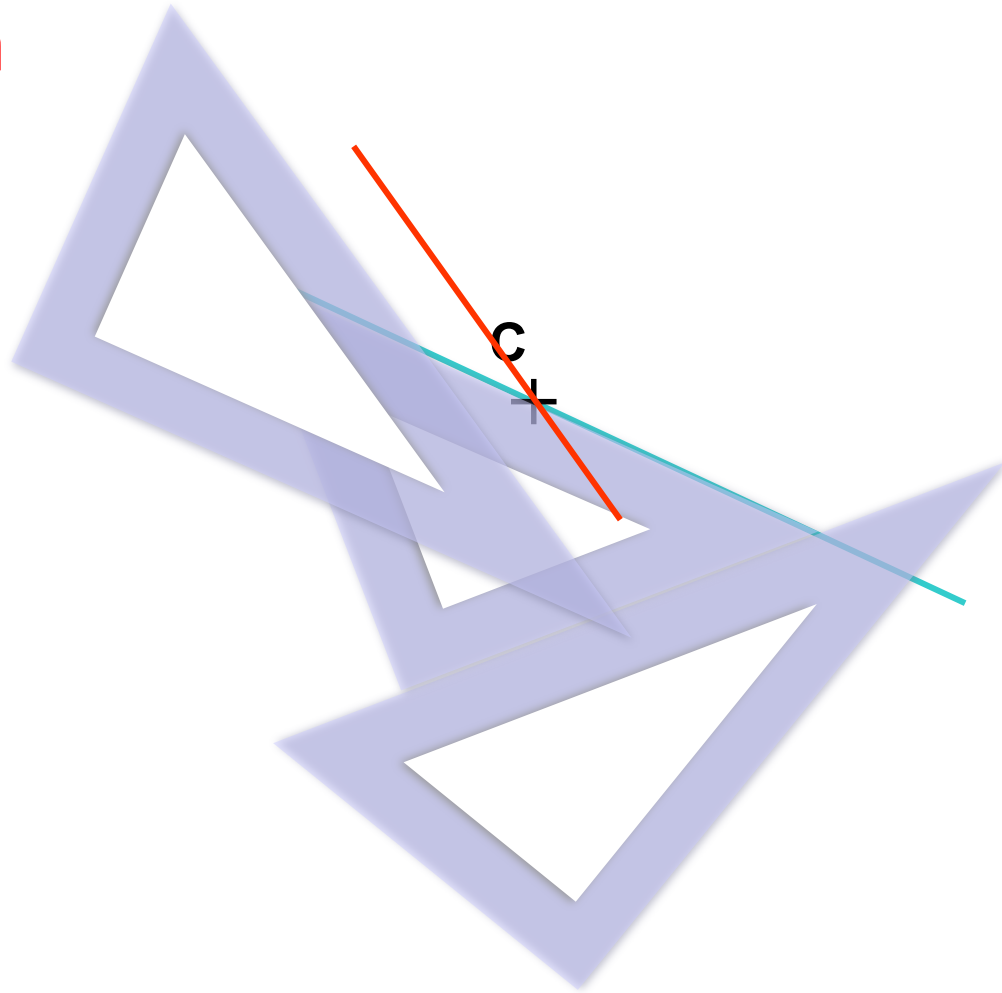
To draw a line making 15° with a given line and pass through a given point.

Given



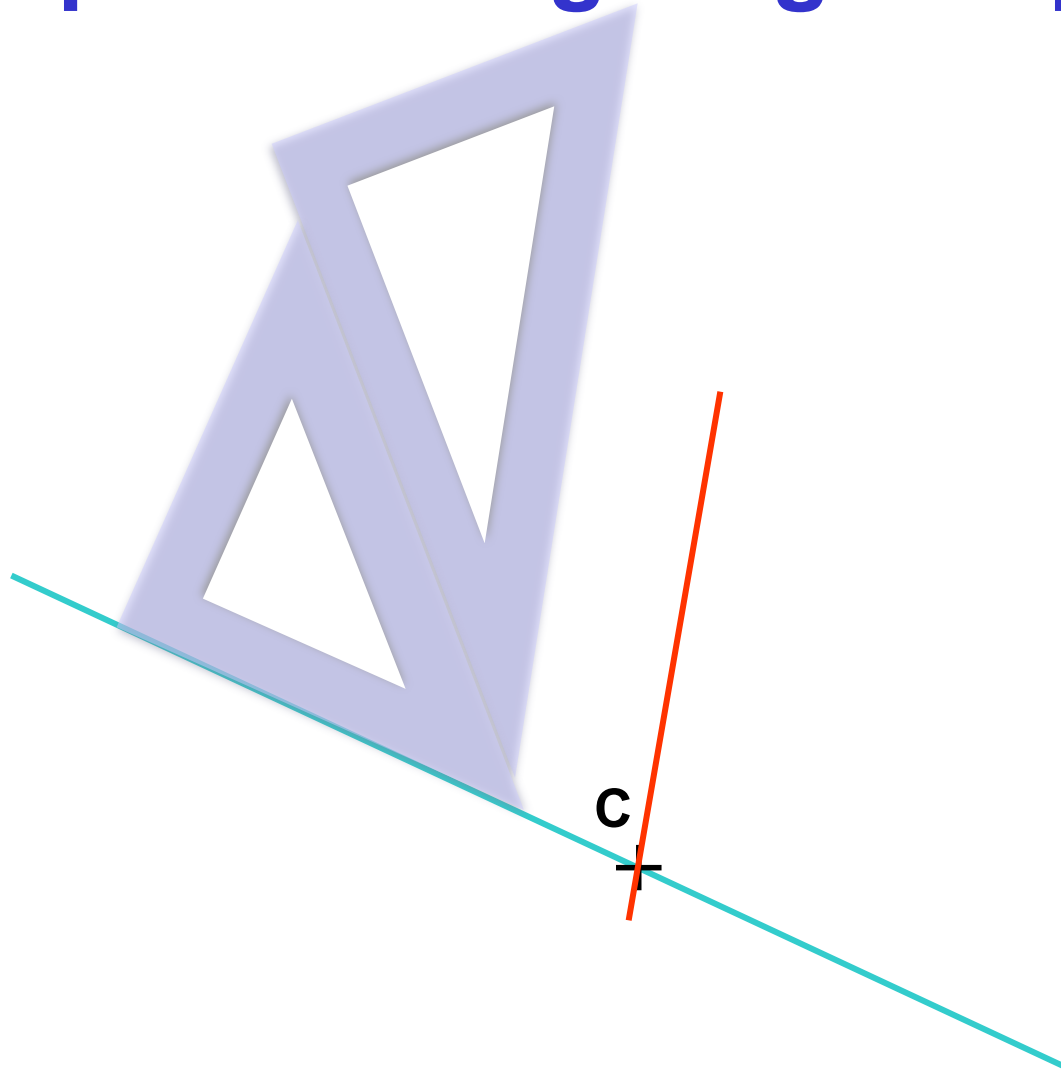
To draw a line making 30° with a given line and pass through a given point.

Given

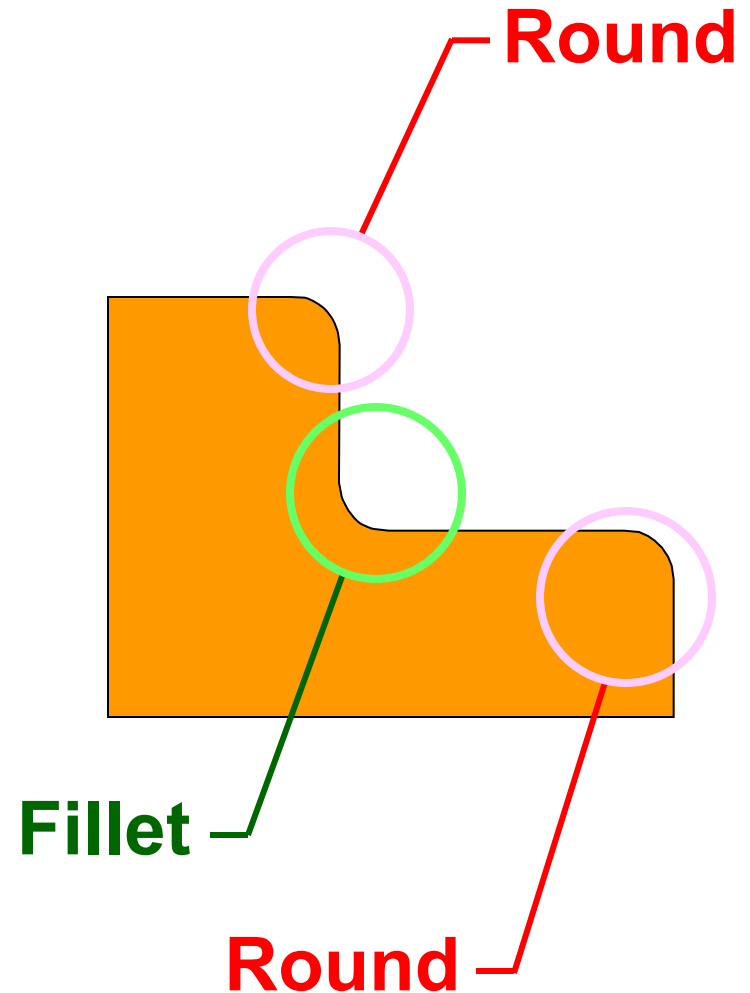
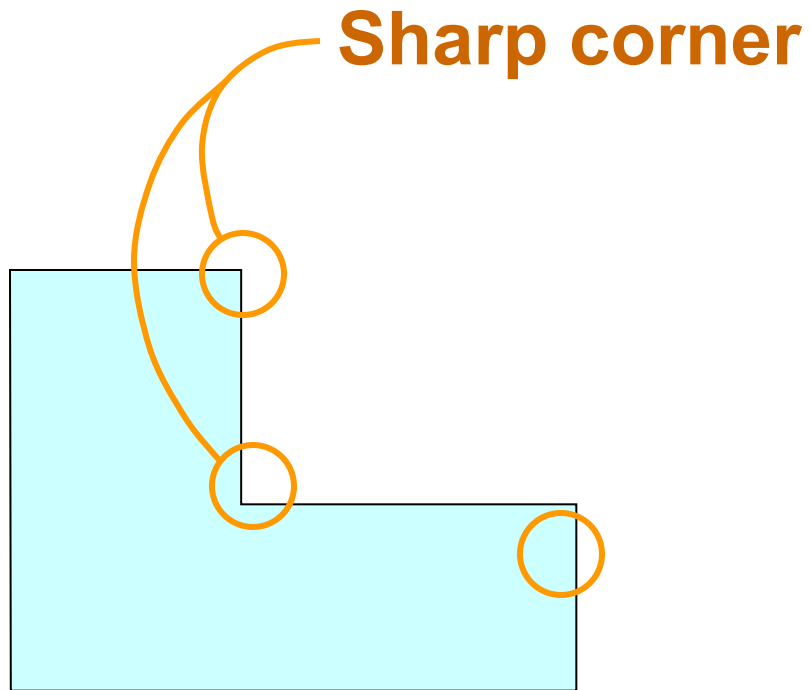


To draw a line making 75° with a given line and pass through a given point.

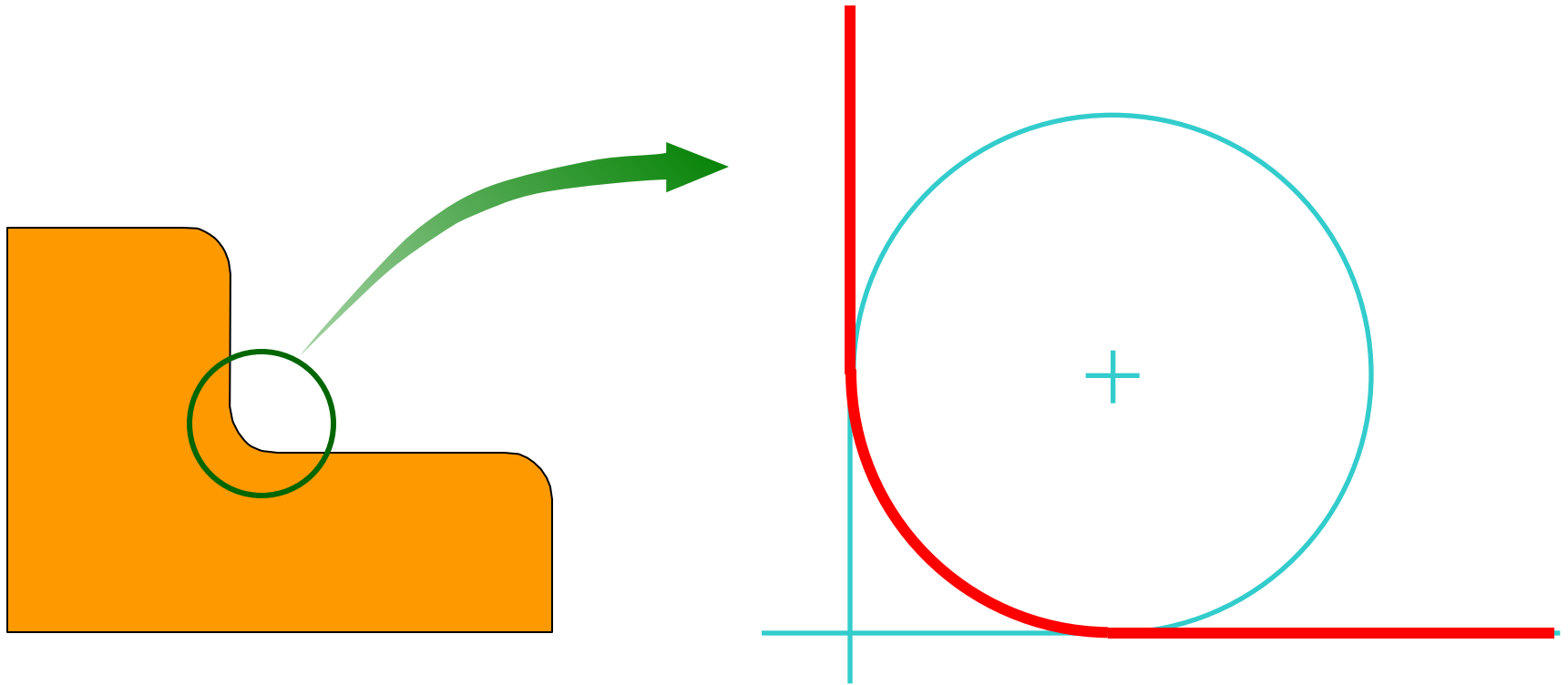
Given



FILLET AND ROUND



FILLET AND ROUND

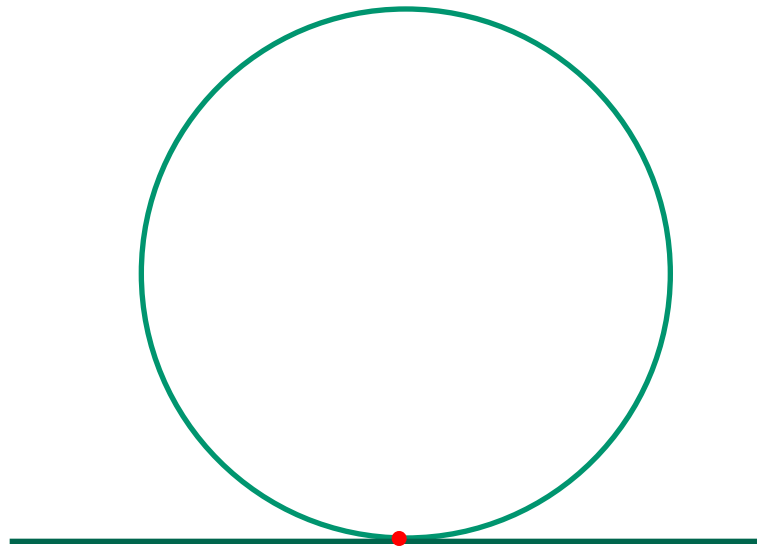


To draw the arc, we must find the location of the center of that arc.

How do we find the center of the arc?

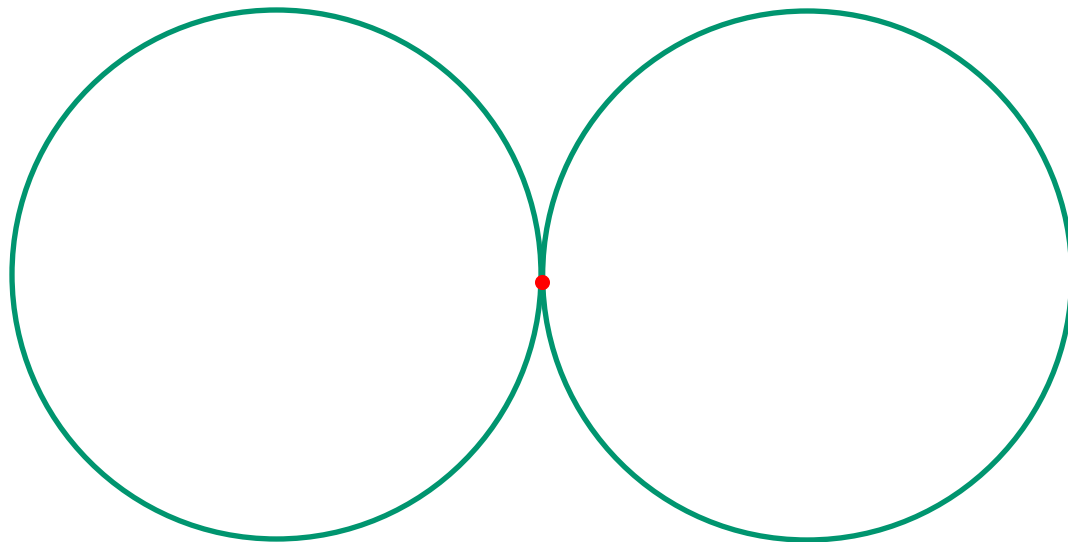
Tangents- Definition

A line is a tangent to a circle if it touches the circle at only one point



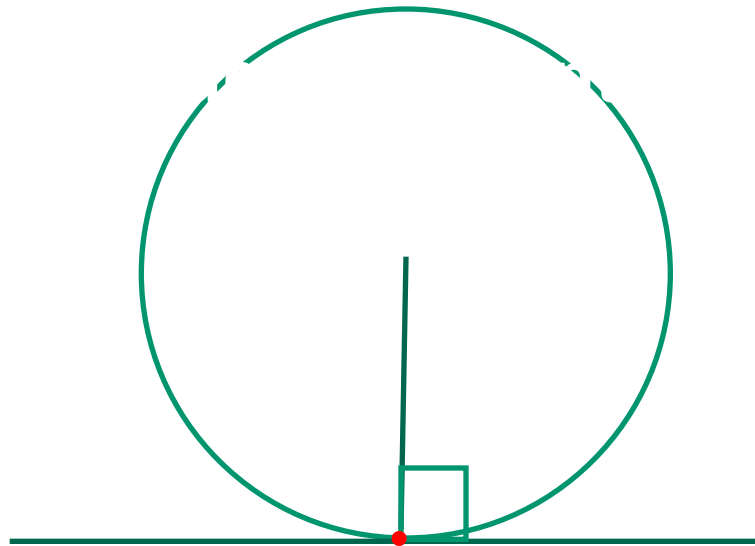
Tangents- Definition

Two curves are tangent to each other if the touch in one and only one place



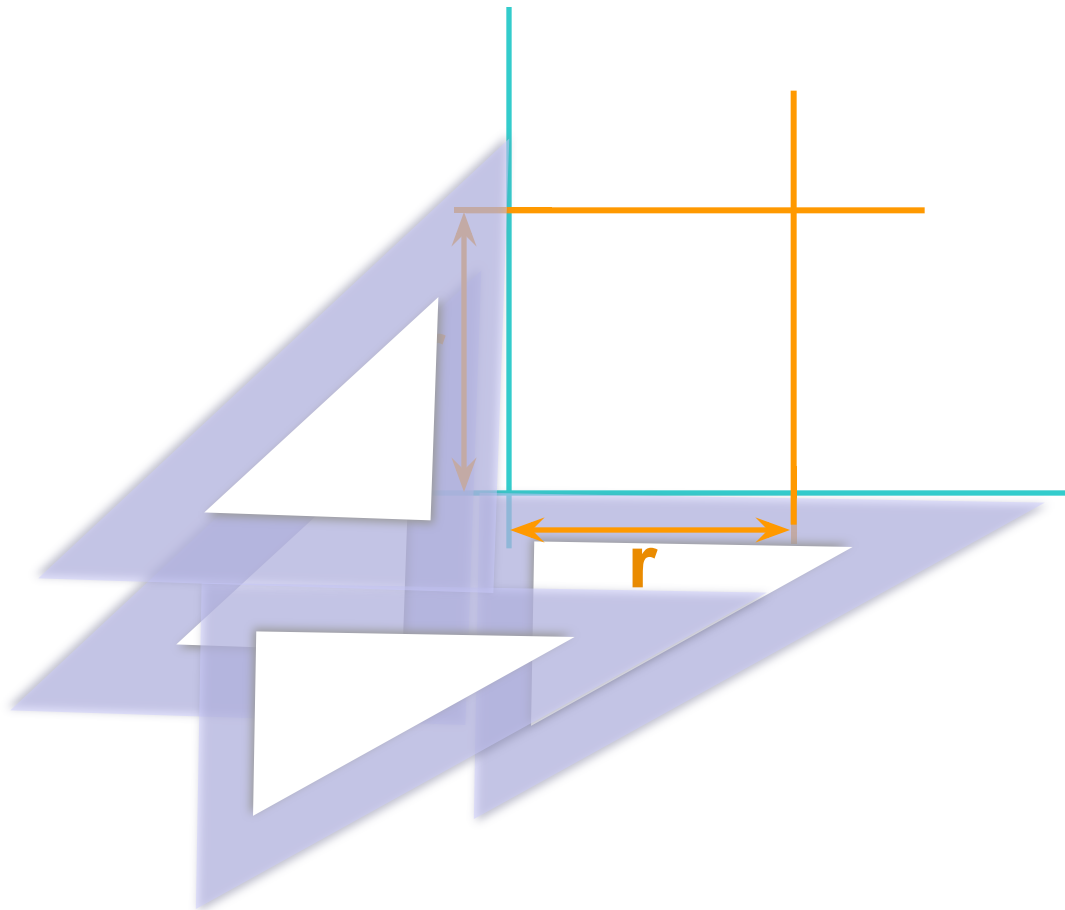
Tangents- Geometric location of tangent point

The tangent point of a straight line and circle will lie at the intersection of a perpendicular to the straight line that passes through the center of the circle.



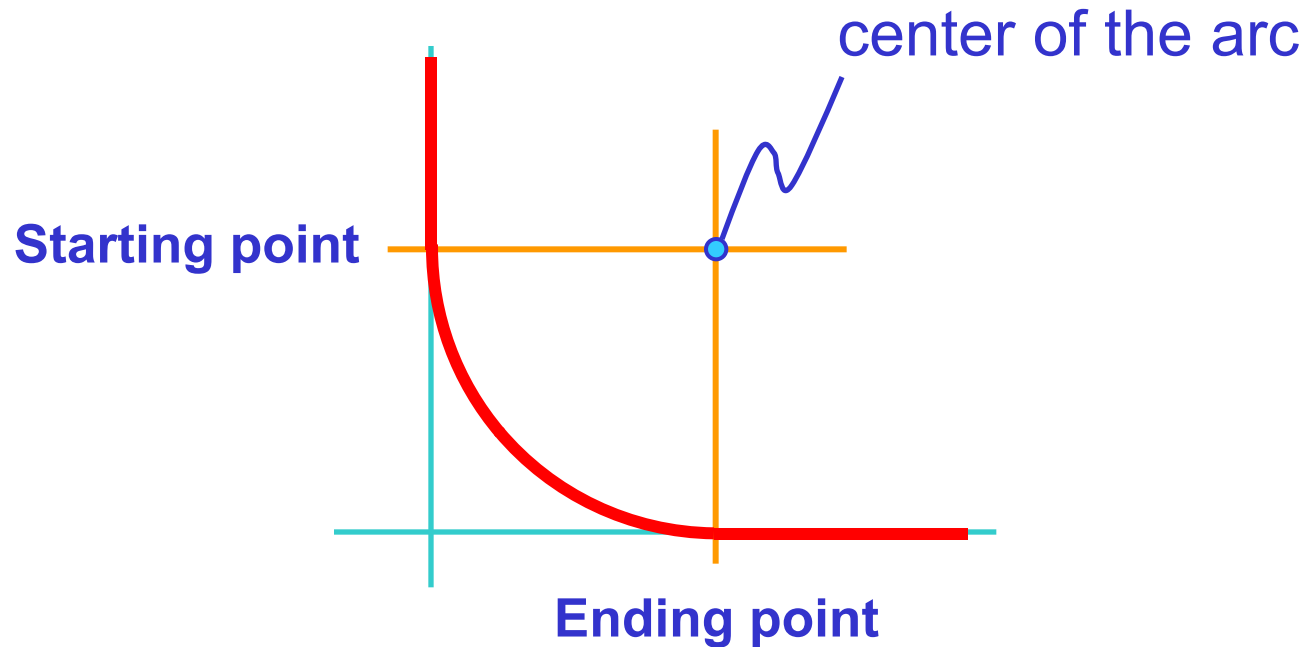
To draw an arc of given radius tangent to two perpendicular lines

Given arc radius r



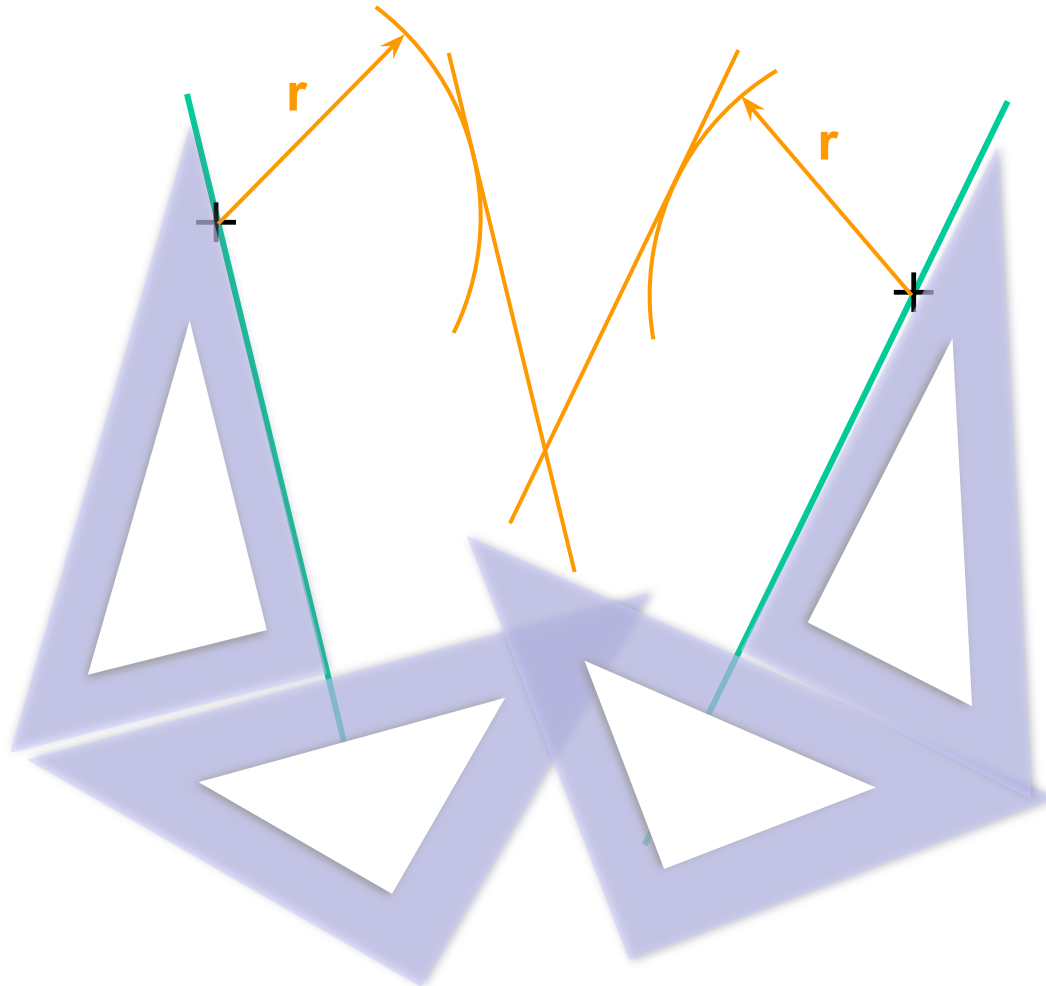
To draw an arc of given radius tangent to two perpendicular lines

Given arc radius r



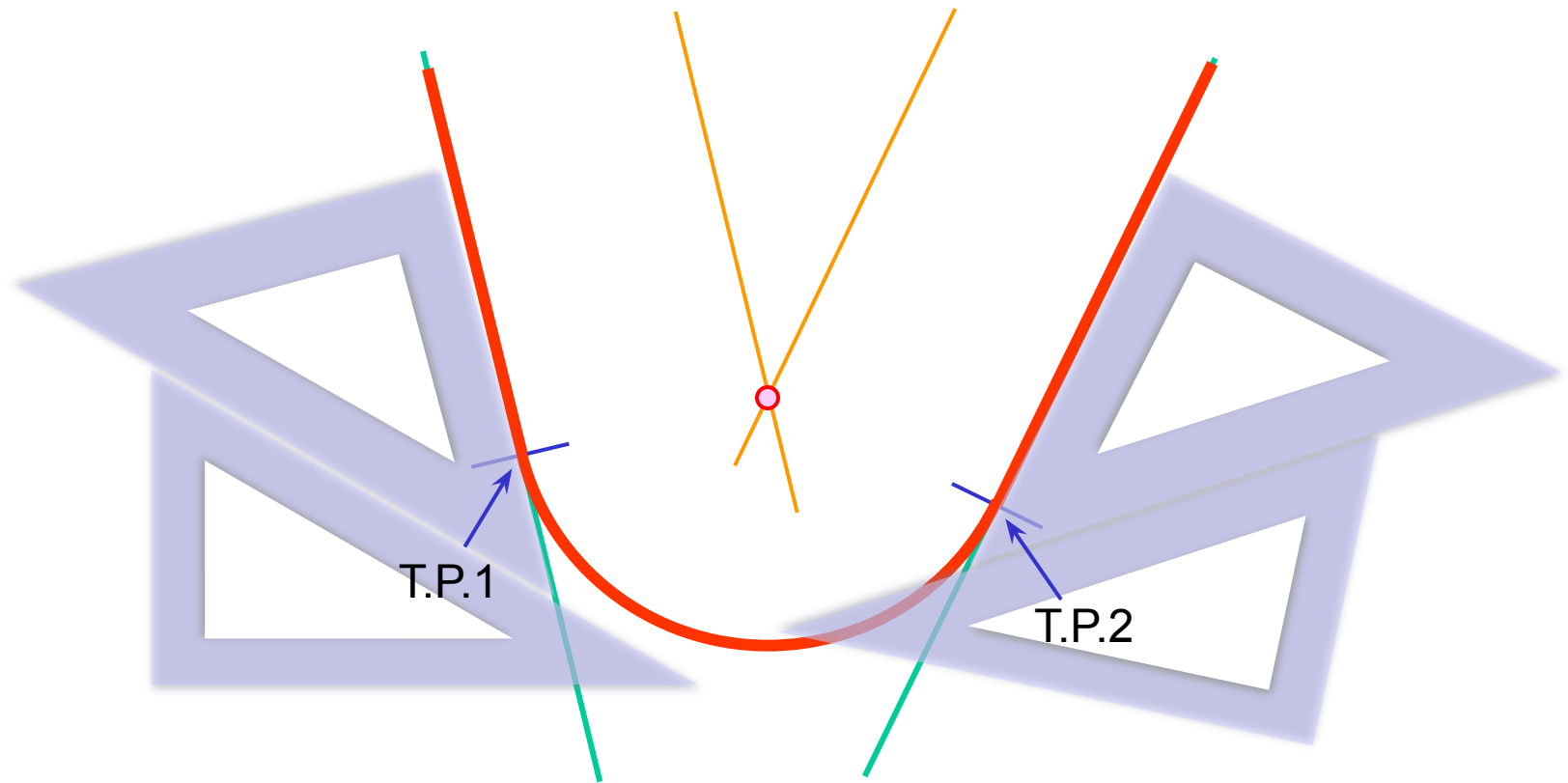
To draw an arc of given radius tangent to two lines

Given arc radius r

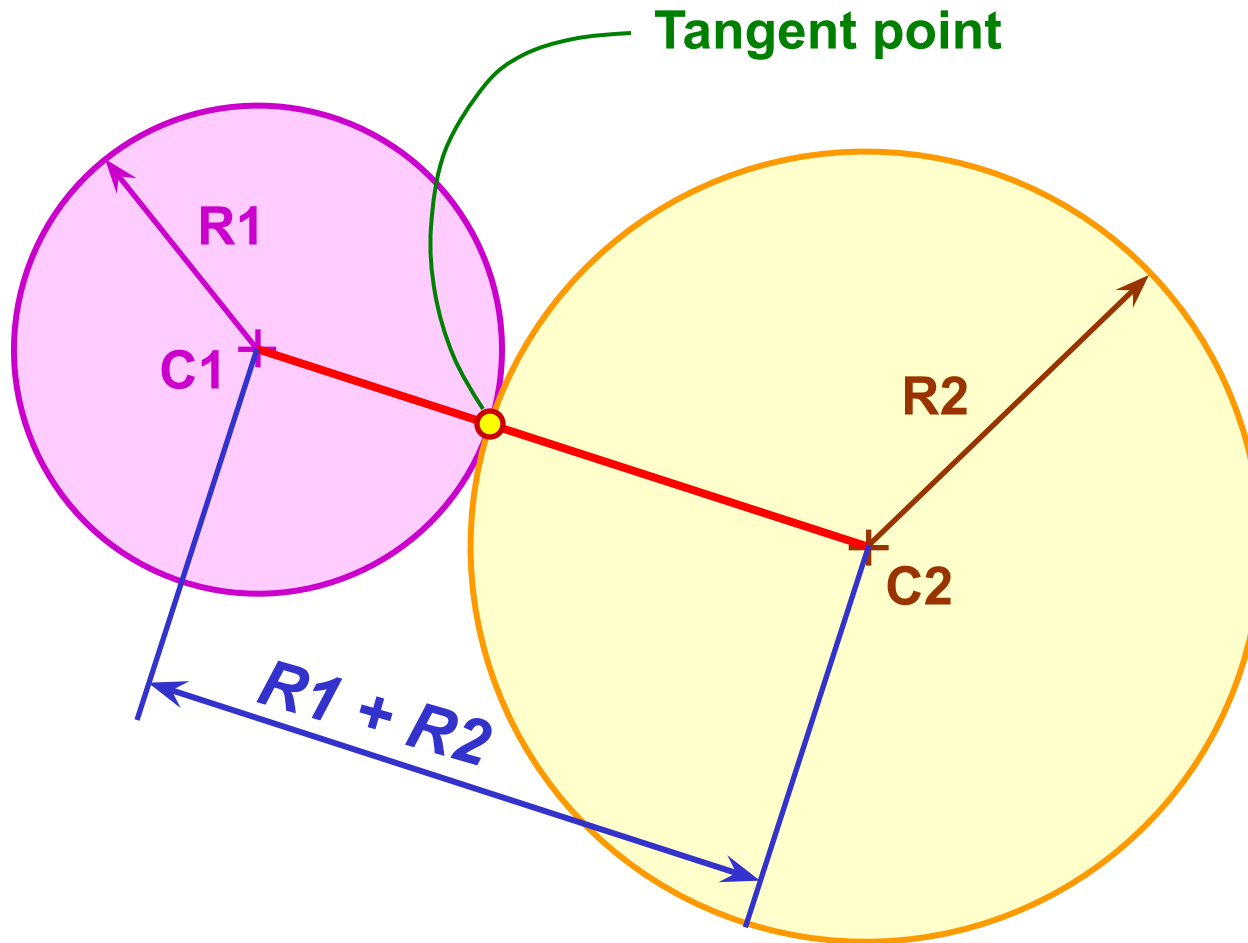


To draw an arc of given radius tangent to two lines

Given arc radius r



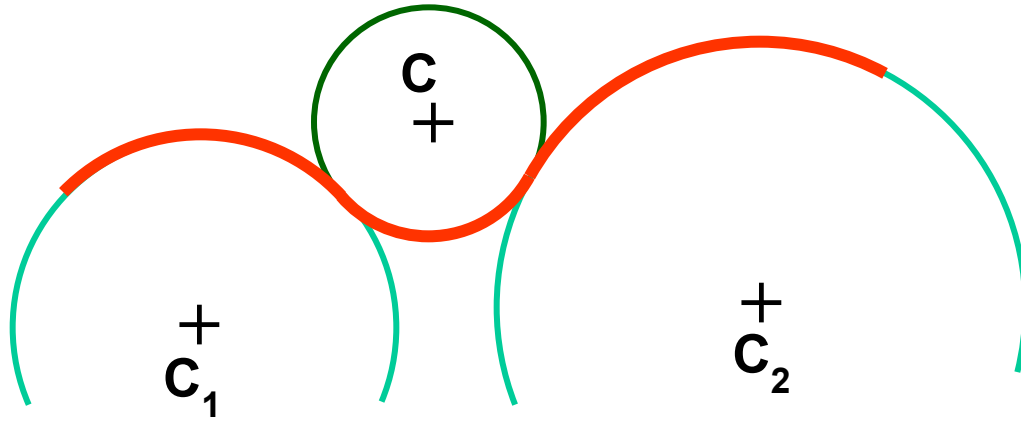
When circle tangent to other circle



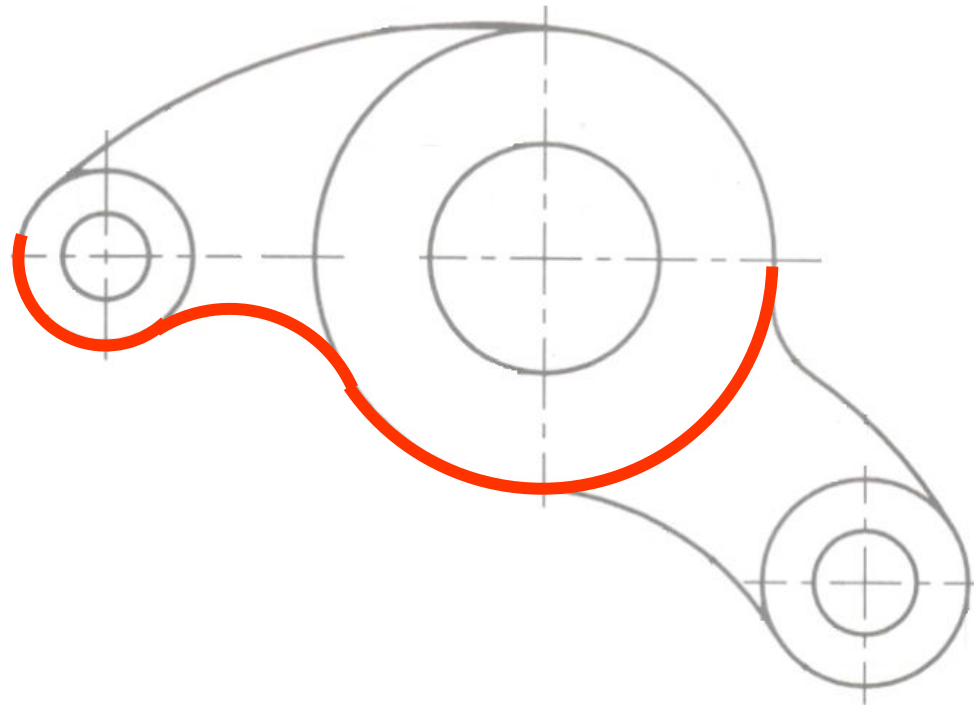
The center of two circles and tangent point lie on the same straight line !!!

To draw a circle tangent to two circles I

Given

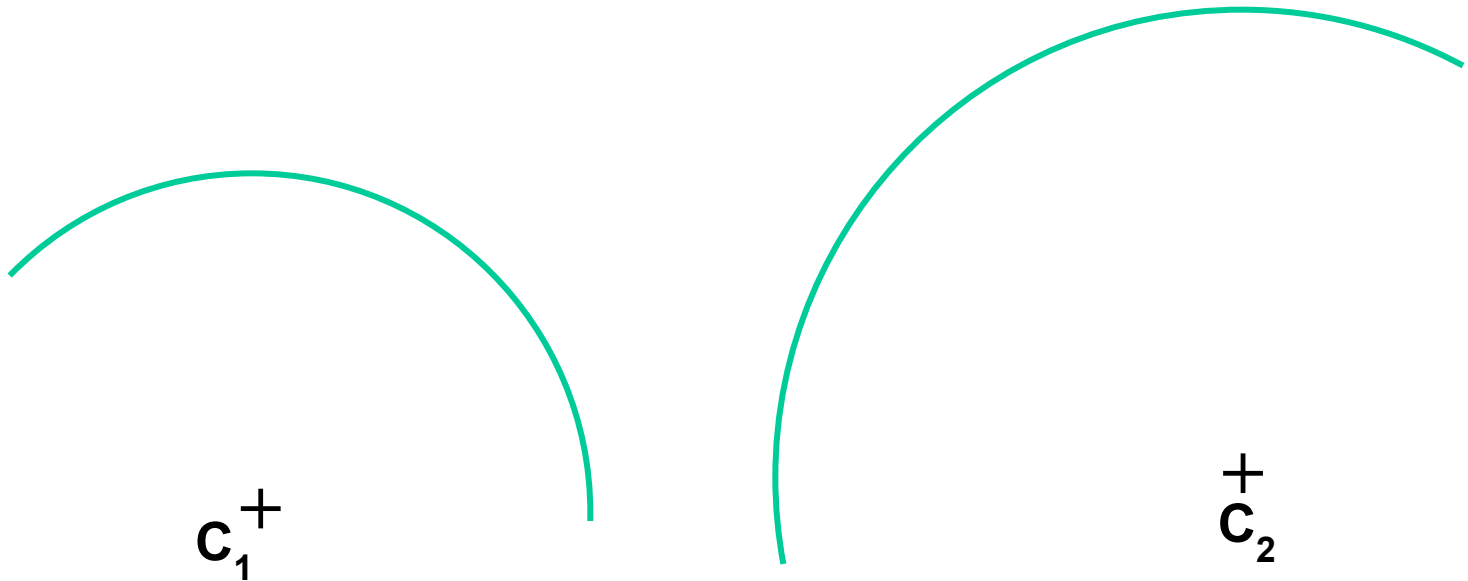


Example



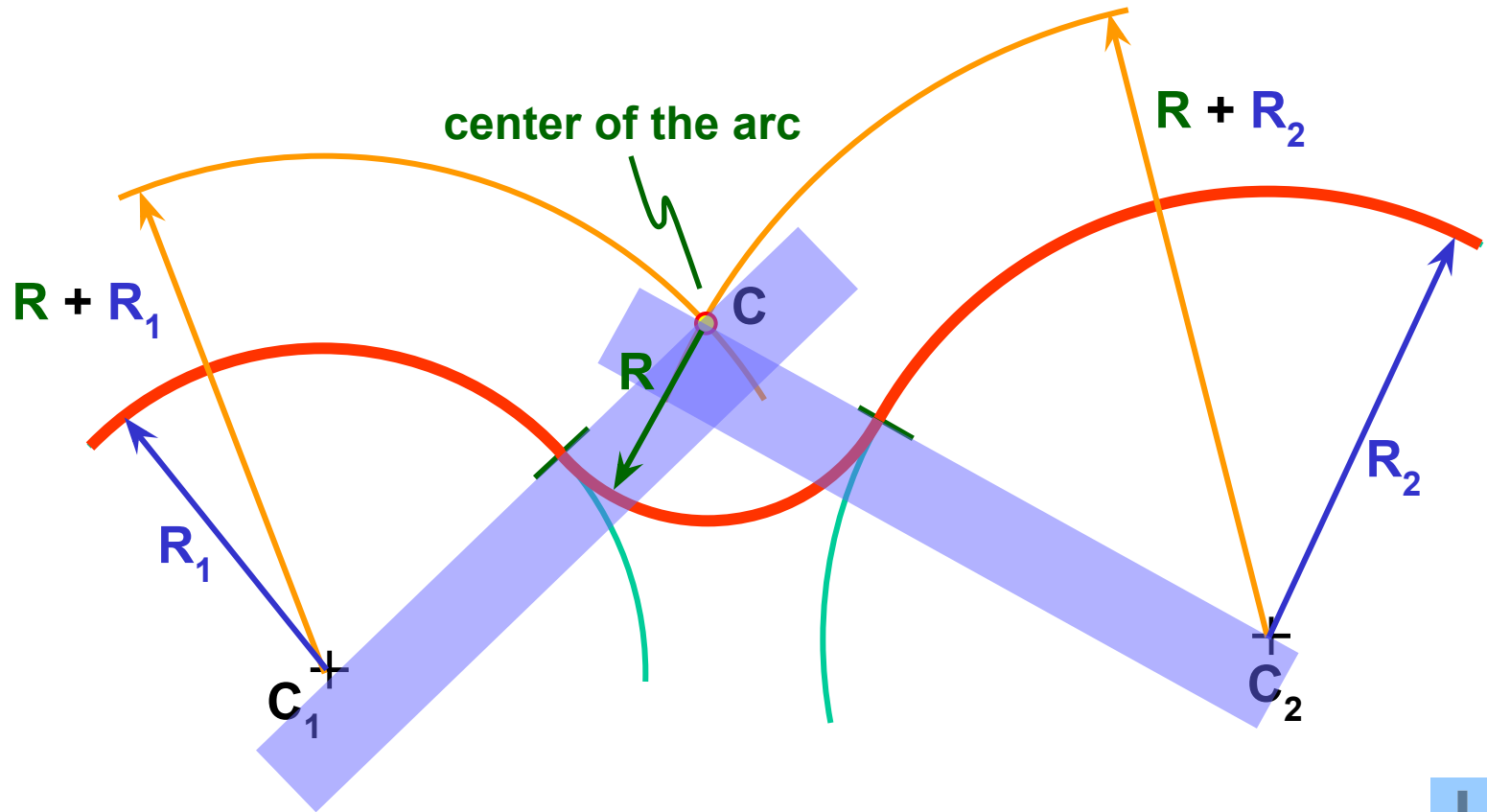
To draw a circle tangent to two circles I

Given Two circles and the radius of the third circle = R



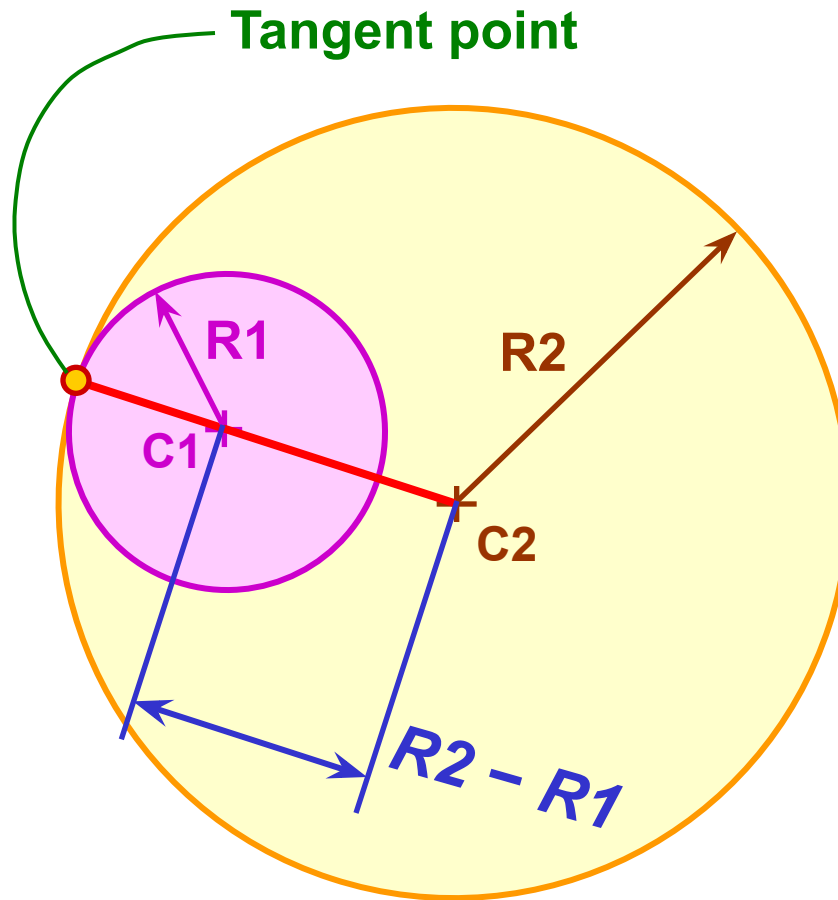
To draw a circle tangent to two circles I

Given Two circles and the radius of the third circle = R



Repeat

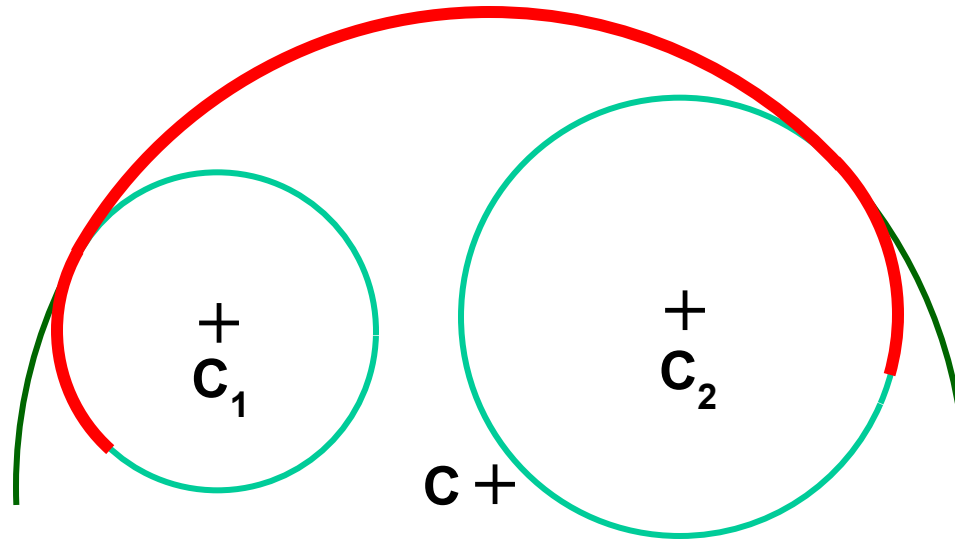
When circle tangent to other circle



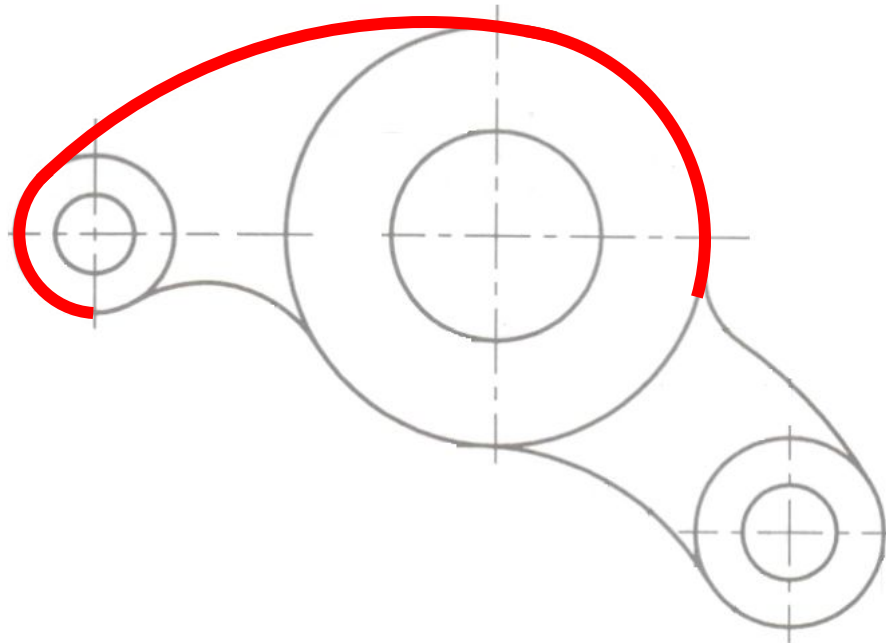
The center of two circles and tangent point must lie on the same straight line !!!

To draw a circle tangent to two circles II

Given

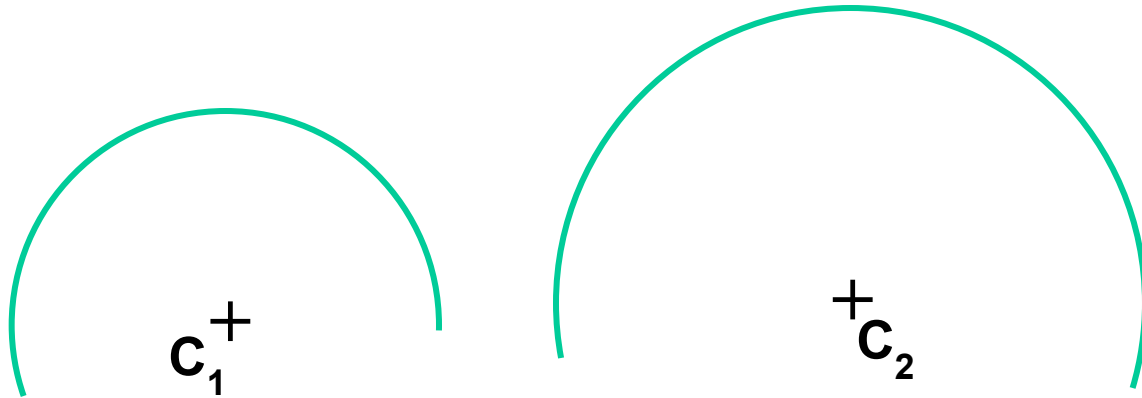


Example



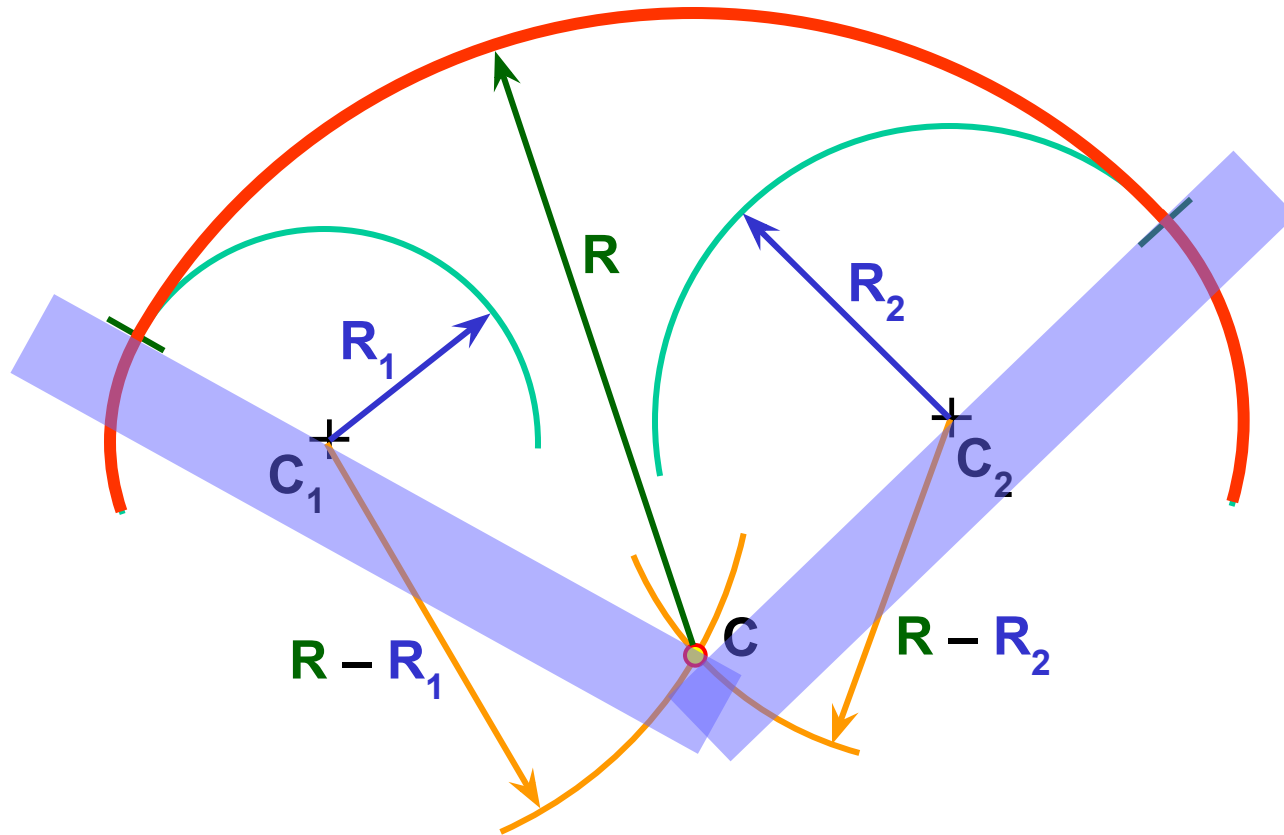
To draw a circle tangent to two circles II

Given Two circles and the radius of the third circle = R



To draw a circle tangent to two circles II

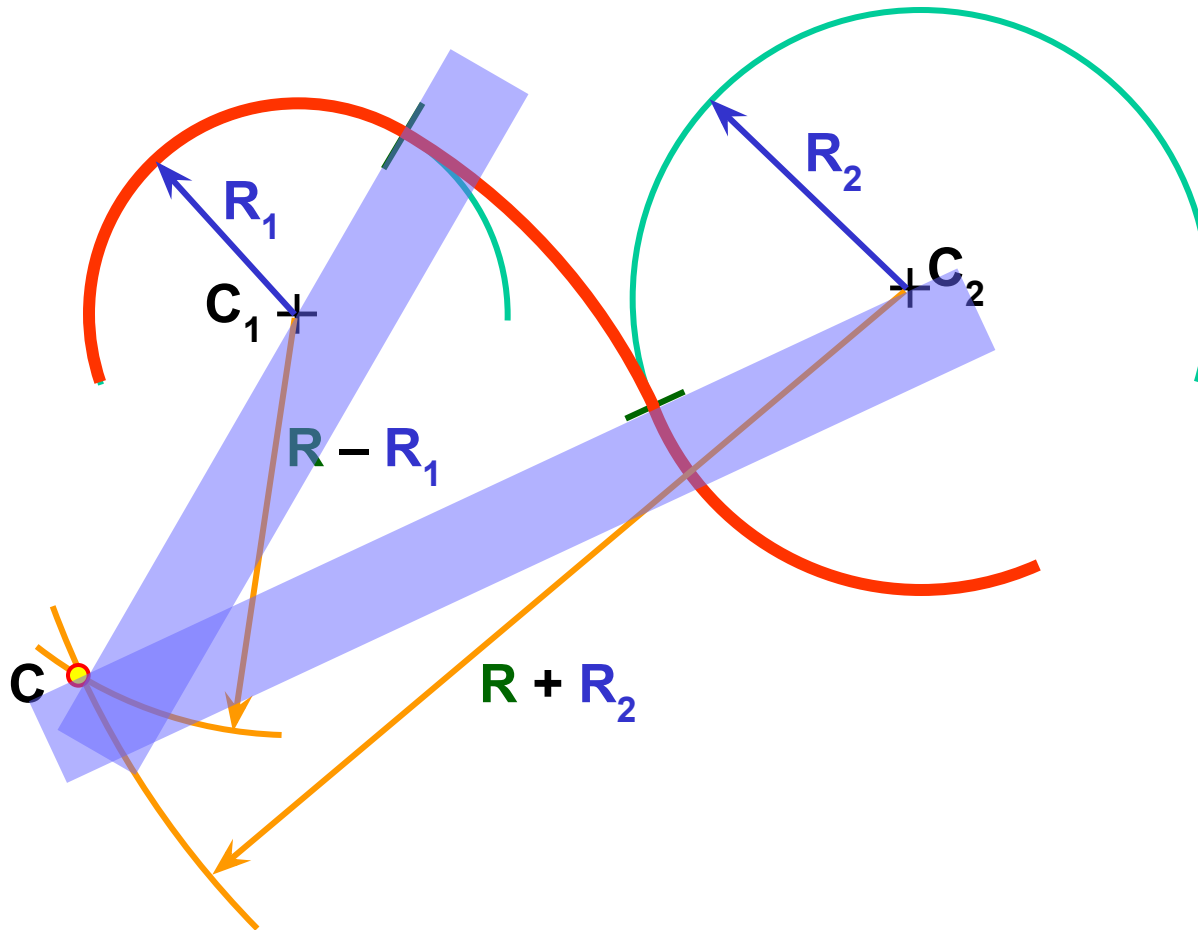
Given Two circles and the radius of the third circle = R



Repeat

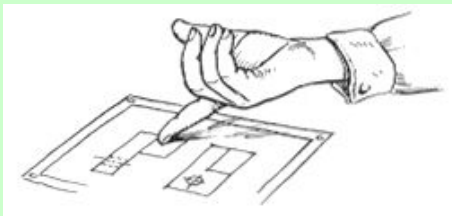
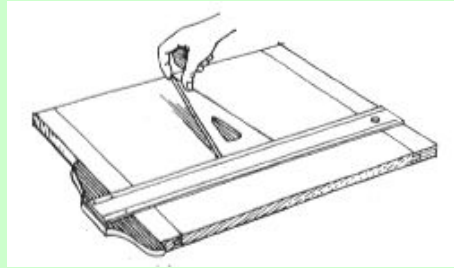
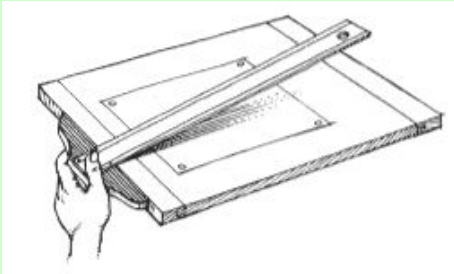
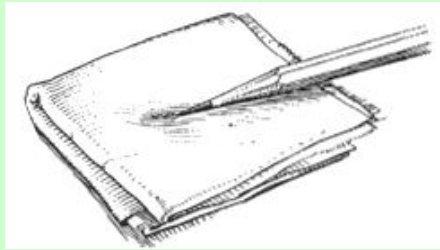
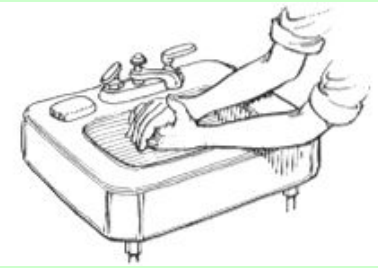
To draw a circle tangent to two circles III

Given Two circles and the radius of the third circle = R



How to Keep Your Drawing Clean

Do



Don't

